INCREASING STABILITY FOR THE INVERSE POTENTIAL PROBLEM OF THE FRACTIONAL HELMHOLTZ EQUATION

DONG QIU, XIANG XU, AND YUE ZHAO

Abstract. This paper is concerned with the inverse potential scattering problem for the three-dimensional fractional Helmholtz equation. For every $0 < \alpha < 1$, we establish an increasing stability of determining the potential function from multifrequency exterior measurements of the scattered field. The stability consists of a Lipschitz data discrepancy and a logarithmic remainder that decreases when the upper measured wavenumber increases. The key ingredient in the analysis is to construct appropriate geometric optics solutions, which relate the X-ray transform of the potential to the measured fields. As a consequence, the stability is derived by applying analytic continuation. Two-dimensional numerical examples with noisy near-field data illustrate the benefit of enlarging the measured frequency band for three representative fractional orders.

1. Introduction

As the basis of fractional quantum mechanics, the fractional Schrödinger equation was originally established by Laskin via the Feynman path-integral formalism for Lévy-flight particles [5, 6]. Concurrently, its associated fractional Laplacian has emerged as a powerful effective model capable of capturing intrinsic non-locality, power-law dispersion, and anomalous attenuation that arise in a variety of physical systems, including quantum optics [16, 17], geophysical electromagnetics [13], and electrodynamics [22]. Driven by the important applications, the fractional Schrödinger equation has been extensively investigated in both mathematical and physical regimes, see e.g., [7, 19, 23, 20] and the references cited therein.

We consider the equation

$$(-\Delta)^{\alpha} u - k^{2\alpha} u + Vu = 0 \quad \text{in } \mathbb{R}^3, \qquad 0 < \alpha < 1, \tag{1.1}$$

where $k > 0$ is the wavenumber and V is a real-valued, compactly supported potential independent of k. We use the Fourier transform convention for the fractional Laplacian operator

$$\widehat{f}(\xi) = \int_{\mathbb{R}^3} e^{-ix\cdot\xi} f(x)\,\mathrm{d}x, \qquad \widehat{(-\Delta)^{\alpha} f}(\xi) = |\xi|^{2\alpha} \widehat{f}(\xi).$$

For the incident plane wave $u^{\mathrm{inc}} = e^{ikx\cdot d}$, $d \in \mathbb{S}^2$, write the total field as $u = u^{\mathrm{inc}} + u^s$. The scattered field is outgoing and satisfies

$$u^s(\rho\theta) = O(\rho^{-1}), \qquad \rho(\partial_\rho - ik)u^s(\rho\theta) \longrightarrow 0 \quad \text{as } \rho \to \infty, \tag{1.2}$$

2020 *Mathematics Subject Classification.* 35R30, 35R11, 35P25.

Key words and phrases. fractional Helmholtz equation, inverse potential problem, increasing stability, near-field data, X-ray transform.

uniformly in $\theta \in \mathbb{S}^2$, where $\rho = |x|$ and $\theta \in \mathbb{S}^2$. The direct problem is to determine the scattered field u^s for a given potential and incident wave, while the inverse problem is to reconstruct V from exterior measurements of u^s.

For the direct scattering problems, a great deal of mathematical and numerical results are available involving the fractional Laplacian, in particular the fractional Schrödinger equation. In quantum optics, bound states, resonances, and related spectral theory for nonlocal fractional partial differential equations have been developed in [17]. High-order numerical methods have been proposed for simulating single-excitation states [16]. Complementary to the results in mathematical theory of the scattering resonances for the classical Schrödinger operator, meromorphic continuation for the resolvent of the fractional Schrödinger operators was investigated in [18], which yields a logarithmic analytic region and resolvent estimates. Fast and scalable numerical methods for addressing the non-locality and high-frequency oscillations of fractional Schrödinger operators are developed in [12, 13], which find applications in wave propagation and electromagnetic modeling.

Inverse problems for the fractional Schrödinger equation have received considerable attention in recent years. Much of the existing work concerns the fractional Calderón problem, which seeks to recover a potential from a nonlocal exterior Dirichlet-to-Neumann map. Uniqueness, stability and instability on this topic were explored in [14, 4, 21]. Compared with the well-studied fractional Calderón problems, little is known for the uniqueness and stability for the inverse scattering problems modeled by the fractional Schrödinger equation. In more recent works, uniqueness results for the inverse source and potential scattering problems were proved by using the far-field pattern at multiple frequencies [2, 7]. Increasing stability for the inverse source problem of the stochastic fractional Helmholtz equation was established in [10]. Bayesian approaches to inverse scattering for fractional Helmholtz models were studied in [15], and the factorization method for support reconstruction was developed in [24]. For related reconstruction methods and Bayesian analysis for the fractional Calderón problem, see also [25, 26, 27, 28]. The present work concerns stability for recovering a deterministic potential from exterior scattered fields.

We formulate the main result of this paper. Fix $0 < r < R$ and $F > 0$, and introduce the a priori class

$$\mathcal{V}^4_{r,F} = \{V \in C^4_c(B_r; \mathbb{R}) : \|V\|_{C^4(\mathbb{R}^3)} \le F\}. \tag{1.3}$$

Here $B_r = \{x : |x| < r\}$ and $\|V\|_{C^4} = \max_{|\nu|\le 4} \|\partial^\nu V\|_\infty$. The observations are made either on an enclosing sphere or in an exterior annulus. Set

$$\Gamma = \partial B_R, \qquad A = B_R \setminus \overline{B_r}, \qquad \mathcal{O} \in \{\Gamma, A\}, \qquad Y = L^2(\mathcal{O} \times \mathbb{S}^2). \tag{1.4}$$

The observation measure is surface measure on Γ or volume measure on A, together with surface measure in d. For two potentials let $u^s_j = u^s_{V_j}$, $j = 1, 2$, and put $\tau = (2\alpha - 1)_+ = \max(2\alpha - 1, 0)$.

The following increasing stability for the inverse potential scattering problem is the main result of this paper. It shows that the stability will improve as the upper bound of the bandwidth of the utilized frequencies increases.

Theorem 1.1. *Let $0 < \alpha < 1$ and $V_1, V_2 \in \mathcal{V}^4_{r,F}$. There are constants $K_0, C > 0$ such that the following holds for $K \ge K_0 + 1$. Define*

$$\varepsilon_w = \sup_{k\in[K_0,K]} k^\tau \left\|u^s_1(k) - u^s_2(k)\right\|_Y. \tag{1.5}$$

If $0 < \varepsilon_w < e^{-1}$, *set* $E = |\log \varepsilon_w|$ *and* $\rho = \min(8\alpha, 4)$. *Then the following stability estimate holds*

$$\|V_1 - V_2\|^2_{H^{-1/2}(\mathbb{R}^3)} \leq C \left(\varepsilon_w^2 + \left(K + \sqrt{\frac{E}{\log(2+E)}} \right)^{-\rho} \right). \tag{1.6}$$

The constants are independent of K *and* ε_w. *As a consequence, an increasing stability estimate also holds for the* L^2 *norm of the potential by an argument of interpolation.*

In this work, we establish high-frequency well-posedness of the direct scattering problem for every $0 < \alpha < 1$, together with uniform resolvent estimates and holomorphic continuation in a fixed complex strip. For the inverse problem, our approach employs the construction of the geometric solutions, which relate the X-ray transform of the potential to the exterior measurements. To do this, we construct an exact real phase and a positive leading amplitude for the geometric solutions. The fractional order determines the size of the leading phase correction, which is proportional to $k^{1-2\alpha}$ times a half-ray integral of the potential. This correction tends to zero for $\alpha > 1/2$, is of order one for $\alpha = 1/2$, and may grow for $\alpha < 1/2$. In the last case, its exponential may wind around the unit circle. To deal with this issue, we use the integer combinations of phases to cancel the second phase coefficient, and the frequency averaging method to control the phase ambiguity for $\alpha < 1/2$. As a consequence, the Fourier transform of the potential can be controlled by the averaged frequency data with a high-frequency remainder term. Then the stability follows from using a quantitative analytic continuation developed by [1]. Finally, we implement a regularized multifrequency reconstruction method for the two-dimensional model. Numerical examples for different fractional orders illustrate the bandwidth dependence of the reconstruction accuracy.

This paper is organized as follows. Section 2 establishes direct scattering and the resolvent estimates. Section 3 constructs the geometric optics approximation and proves its error estimates in the volume and on the receiver sphere. Section 4 relates the phases to ray integrals and proves Theorem 1.1 by frequency combinations and analytic continuation. Section 5 presents the reconstruction method and numerical examples. Section 6 concludes the paper, and the appendices provide some technical proofs.

Throughout the paper, $C > 0$ denotes a generic constant that may change from line to line. Unless otherwise stated, it depends only on the fixed fractional order, geometry, and a priori bound. We also denote $\langle x \rangle = (1 + |x|^2)^{1/2}$, $D = -i\nabla$, $t_- = \max(-t, 0)$, and $\langle f, g \rangle = \int f \overline{g}$.

2. Direct scattering problem

We establish high-frequency well-posedness and the resolvent bounds needed for the field approximation and analytic continuation. As we are focused on the inverse problem, proofs for some results in this section are removed to the appendix, which are related closely to the scattering theory. We refer to the monograph [3] for a comprehensive introduction to the scattering theory.

2.1. **Green function and well-posedness.**

Denote the outgoing free resolvents by

$$\mathcal{R}_{\alpha,0}(k) = \left((-\Delta)^\alpha - k^{2\alpha} - i0 \right)^{-1}, \qquad R_0(k) = (-\Delta - k^2 - i0)^{-1}.$$

In the right half-plane, complex powers use $-\pi/2 < \arg z < \pi/2$. For the free equation, we work with tempered distributions whose Fourier transforms are locally integrable near the origin. The fractional multiplier is defined by multiplication there and by its smooth action on distributions away from the origin. Outgoing source-generated fields additionally satisfy (1.2). The following decomposition and estimates are proved in Appendix A.1.

Lemma 2.1. *Let $0 < \alpha < 1$. The outgoing free Green function in three dimensions is*

$$G_{\alpha,k}(x) = \frac{k^{2-2\alpha}}{\alpha}\frac{e^{ik|x|}}{4\pi|x|} + J_{\alpha,k}(|x|), \tag{2.1}$$

where

$$J_{\alpha,k}(a) = \frac{\sin(\pi\alpha)}{4\pi^2 a}\int_0^\infty \frac{\gamma^\alpha e^{-a\sqrt{\gamma}}}{\gamma^{2\alpha} - 2\gamma^\alpha k^{2\alpha}\cos(\pi\alpha) + k^{4\alpha}}\,\mathrm{d}\gamma. \tag{2.2}$$

For smooth compactly supported cutoffs χ_1, χ_2, $\chi_1\mathcal{R}_{\alpha,0}(z)\chi_2$ is holomorphic from L^2 to $H^{2\alpha}$ on $\Re z > 0$. For $|\arg z| \le \theta_0 < \pi/2$, $|z| \ge 1$, and $0 \le l \le 2\alpha$,

$$\|\chi_1\mathcal{R}_{\alpha,0}(z)\chi_2\|_{L^2\to H^l} \le C|z|^{1+l-2\alpha}e^{T(\Im z)_-}, \tag{2.3}$$

where $T > 0$ depends on the cutoffs. If $J_\alpha(z)$ denotes the convolution operator associated with the second term, then

$$\|J_\alpha(z)\|_{L^2\to H^l} \le C|z|^{l-2\alpha}. \tag{2.4}$$

For each fixed $k > 0$, $G_{\alpha,k} * f$ is the unique outgoing solution of the free equation with a compactly supported L^2 source f, and belongs to $H^{2\alpha}_{\mathrm{loc}}(\mathbb{R}^3)$.

The Lippmann–Schwinger equation associated with (1.1) is

$$u = e^{ikx\cdot d} - \mathcal{R}_{\alpha,0}(k)(Vu), \qquad u^s = -\mathcal{R}_{\alpha,0}(k)(Vu). \tag{2.5}$$

By Lemma 2.1, this integral equation is equivalent to (1.1) with the radiation condition (1.2) in the distributional class specified above.

We denote the outgoing perturbed resolvent by

$$\mathcal{R}_{\alpha,V}(z) = \left((-\Delta)^\alpha + V - z^{2\alpha}\right)^{-1}_{\mathrm{out}}.$$

For z in the first quadrant, it agrees with the spectral resolvent. Elsewhere, the localized resolvent is defined by outgoing meromorphic continuation.

Theorem 2.2. *Let $0 < \alpha < 1$ and $V \in L^\infty_c(\mathbb{R}^3;\mathbb{R})$. There is a discrete set $\Sigma_V \subset (0,\infty)$ such that, for $k \notin \Sigma_V$, (2.5) has a unique solution with $u^s \in H^{2\alpha}_{\mathrm{loc}}(\mathbb{R}^3)$. The scattered field depends continuously in $H^{2\alpha}_{\mathrm{loc}}$ on the L^2 restriction of the incident field to a fixed neighborhood of* $\operatorname{supp} V$. *The localized outgoing resolvent is meromorphic on $\Re z > 0$.*

If in addition $V \in C^1_c(\mathbb{R}^3;\mathbb{R})$, then

$$\Sigma_V \subset \{k > 0 : k^{2\alpha} \le E_*(V)\}, \tag{2.6}$$

where

$$E_*(V) = \max\left(0, \sup_{x\in\mathbb{R}^3}\left(V(x) + \frac{x\cdot\nabla V(x)}{2\alpha}\right)\right). \tag{2.7}$$

In particular, the direct problem is well posed at every sufficiently large frequency for every $0 < \alpha < 1$.

The proof for this theorem provided in Appendix A.2 uses analytic Fredholm theory to construct the outgoing solution. At an exceptional real frequency, the corresponding homogeneous solution is an eigenfunction; a scaling identity gives (2.6).

Remark 2.3. For potentials supported in B_r with a common C^1 bound, the threshold is uniform, since

$$E_*(V) \le \|V\|_\infty + \frac{r}{2\alpha}\|\nabla V\|_\infty .$$

2.2. **High-frequency resolvent estimates.** For the remainder of the paper, the potentials belong to $\mathcal{V}^4_{r,F}$. Put $\mu = 2\alpha$. The following weighted estimate is proved by a positive commutator argument in Appendix A.3.

Lemma 2.4. *Let $0 < \mu < 2$ and $V \in \mathcal{V}^4_{r,F}$. There are $k_0, C > 0$ such that the outgoing resolvent satisfies*

$$\left\|\langle x\rangle^{-1}\mathcal{R}_{\alpha,V}(k)\langle x\rangle^{-1}\right\|_{L^2\to L^2} \le Ck^{1-\mu}, \qquad k \ge k_0. \tag{2.8}$$

The same estimate holds with two fixed compactly supported cutoffs in place of the spatial weights.

The free-resolvent decomposition gives a fixed strip of holomorphy for the localized perturbed resolvent; see Appendix A.4.

Proposition 2.5. *Let $0 < \mu < 2$ and fix compactly supported smooth cutoffs η_1, η_2. There exist $k_0, b, C > 0$ such that, uniformly for $V \in \mathcal{V}^4_{r,F}$, the localized outgoing resolvent $\eta_1\mathcal{R}_{\alpha,V}(z)\eta_2$ is holomorphic in*

$$\mathcal{S}_{k_0,b} = \{z \in \mathbb{C} : \Re z > k_0,\ |\Im z| < b\}. \tag{2.9}$$

Moreover,

$$\|\eta_1\mathcal{R}_{\alpha,V}(z)\eta_2\|_{L^2\to L^2} \le C(\Re z)^{1-\mu}, \qquad z \in \mathcal{S}_{k_0,b}. \tag{2.10}$$

Corollary 2.6. *Let Y be defined by* (1.4)*. Uniformly for $V \in \mathcal{V}^4_{r,F}$, the map $z \mapsto u^s_V(\cdot,\cdot,z)$ is Y-valued holomorphic in a strip of the form* (2.9)*, and*

$$\|u^s_V(\cdot,\cdot,z)\|_Y \le C(1+|z|)^3. \tag{2.11}$$

The total field and its modulation by $e^{-izx\cdot d}$ have the same properties.

The volume estimate follows from Proposition 2.5. For spherical observations, the separation of the receivers from the potential support gives a holomorphic integral kernel; see Appendix A.5.

3. Geometric optics approximation

In this section, we investigate the construction of geometric solution to the fractional Helmholtz equation, which is useful in the subsequent analysis of the inverse problem. To do this, we construct a real phase and a positive leading amplitude, then estimate the difference between the resulting approximation and the outgoing field. Throughout this section, $\mu = 2\alpha$ and Y is defined by (1.4).

For a compactly supported function f, define its half-ray and X-ray transforms by

$$Bf(x,d) = \int_{-\infty}^{0} f(x+sd)\,\mathrm{d}s, \qquad Xf(x,d) = \int_{\mathbb{R}} f(x+sd)\,\mathrm{d}s. \tag{3.1}$$

They satisfy $d \cdot \nabla Bf = f$ and $Xf(x,d) = Bf(x,d) + Bf(x,-d)$. Put $h = k^{-1}$, $\delta = h^\mu$, and

$$P_{h,V} = |hD|^\mu - 1 + \delta V, \qquad R_{h,V} = (P_{h,V} - i0)^{-1}.$$

Rescaling Lemma 2.4 gives

$$\left\| \langle x \rangle^{-1} R_{h,V} \langle x \rangle^{-1} \right\|_{L^2 \to L^2} \leq C h^{-1}. \tag{3.2}$$

3.1. **The phase and leading amplitude.** The incoming Hamilton flow defines the phase through its action and a positive leading amplitude through the transport equation.

Lemma 3.1. *Let $0 < \mu < 2$ and $V \in \mathcal{V}^4_{r,F}$. There are $\delta_0, C > 0$, depending only on μ, r, R, F, such that for $|\delta| \leq \delta_0$ there are an incoming phase $S_{\delta,V,d}$ and a leading amplitude $a_{0,\delta,V,d}$ with the following properties. The phase is real, and both $S_{\delta,V,d}$ and its spatial gradient have uniformly bounded derivatives through order three jointly in (x,d,δ) on the fixed construction sets. The leading amplitude is real and positive. On a fixed neighborhood of $\overline{B_R}$,*

$$|\nabla S_{\delta,V,d}|^\mu = 1 - \delta V, \qquad \nabla S_{\delta,V,d} = d + O(\delta), \qquad a_{0,\delta,V,d} = 1 + O(\delta) \geq \tfrac{3}{4}. \tag{3.3}$$

Moreover,

$$\begin{aligned} S_{\delta,V,d} - x \cdot d &= -\frac{\delta}{\mu} BV(x,d) + O(\delta^2), \\ \partial_\delta S_{\delta,V,d} &= -\frac{1}{\mu} BV(x,d) + O(\delta). \end{aligned} \tag{3.4}$$

These remainders and the amplitude remainder are uniform in the C^1 norm in the spatial and incident-direction variables. The phase and leading amplitude equal $x \cdot d$ and one in the incoming region $x \cdot d < -r$.

Proof. Treat δ as an independent parameter and prescribe $X(0) = y - L_0 d$, $\Xi(0) = d$ on an incoming section, with $y \in d^\perp$ and $L_0 > r$ fixed. The Hamilton equations are

$$\dot{X} = \mu |\Xi|^{\mu-2} \Xi, \qquad \dot{\Xi} = -\delta \nabla V(X).$$

The Hamiltonian $H_\delta = |\Xi|^\mu + \delta V(X)$ equals one along every trajectory. For sufficiently small $|\delta|$, the momentum remains in $1/2 < |\Xi| < 2$ on the fixed time interval. The vector field and its flow have uniform joint C^3 bounds. The projection $(y,t) \mapsto X(t;y,d,\delta)$ is a small C^1 perturbation of the affine map $y - L_0 d + \mu t d$. It has an inverse on the fixed construction region, after enlarging the parameter sets by a fixed margin. The inverse time and incoming label $y = y(x,d,\delta)$ are jointly C^3. Finitely many transverse frames cover the incident sphere.

Define the reduced action by

$$S_{\delta,d}(x) = -L_0 + \int_0^{t(x,d,\delta)} \Xi(s) \cdot \dot{X}(s) \, \mathrm{d}s.$$

Differentiating along a family of trajectories at fixed δ and integrating by parts give

$$\mathrm{d}S = \Xi(t) \cdot \mathrm{d}x - d \cdot \mathrm{d}X(0) + \int_0^t \mathrm{d}H_\delta(X(s), \Xi(s)) \, \mathrm{d}s.$$

The terminal-time variation is included in $\mathrm{d}x$; the last integral vanishes because every trajectory has energy one. For fixed d, the initial variation satisfies $d \cdot \mathrm{d}y = 0$. For fixed x and a tangential variation of d, use $\mathrm{d}X(0) = \mathrm{d}y - L_0 \, \mathrm{d}d$ and $d \cdot \mathrm{d}y = -y \cdot \mathrm{d}d$. Thus

$$\nabla_x S = \Xi, \qquad \nabla_{\mathbb{S}^2, d} S = y, \qquad S, \Xi, y \in C^3_{x,d,\delta}. \tag{3.5}$$

The intrinsic spherical gradient y lies in $d^\perp$. The identities

$$\Xi(t) - d = -\delta \int_0^t \nabla V(X(s))\,\mathrm{d}s, \qquad S - x\cdot d = \int_0^{t(x,d,\delta)} (\Xi(s) - d)\cdot \dot{X}(s)\,\mathrm{d}s$$

give $\Xi - d = O_{C^3_{x,d}}(\delta)$ and $S - x\cdot d = O_{C^3_{x,d}}(\delta)$ at fixed δ.

We also need the mixed bound

$$\sup_{|\delta|\le\delta_0} \left\|\partial_\delta^3 S\right\|_{C^1_{x,d}} \le C. \tag{3.6}$$

The joint C^3 bound gives a continuous third parameter derivative. For a fixed spatial path γ, integrate the first identity in (3.5) along γ and differentiate three times in δ. Since Ξ is jointly C^3, this gives $\nabla_x \partial_\delta^3 S = \partial_\delta^3 \Xi$. The same argument along paths in a spherical chart, using the incoming label, gives $\nabla_{\mathbb{S}^2,d}\partial_\delta^3 S = \partial_\delta^3 y$. Both right-hand sides are continuous and uniformly bounded, proving (3.6). Differentiating the eikonal equation at zero yields $\partial_\delta S_0 = -BV/\mu$, and Taylor's formula proves (3.4).

Choose a smooth real radial symbol p_{sm} which is zero near the origin and equals $|\xi|^\mu$ for $|\xi| \ge 1/4$. Put $v = \nabla_\xi p_{\mathrm{sm}}(\Xi)$ and solve

$$v\cdot\nabla a_0 + \tfrac{1}{2}(\operatorname{div} v)a_0 = 0, \qquad a_0 = 1 \quad \text{upstream.} \tag{3.7}$$

Here v is jointly C^3 and $\operatorname{div} v$ is jointly C^2. The characteristic formula

$$a_0(x,d,\delta) = \exp\left(-\tfrac{1}{2}\int_0^{t(x,d,\delta)} \operatorname{div} v(X(s),d,\delta)\,\mathrm{d}s\right)$$

gives a joint C^2 bound and, after reducing δ_0, the following estimates at fixed δ:

$$\Xi - d = O_{C^3_{x,d}}(\delta), \qquad a_0 - 1 = O_{C^2_{x,d}}(\delta), \qquad a_0 \ge \tfrac{3}{4}. \tag{3.8}$$

□

3.2. **Volume approximation.** We use the following free propagation estimate for sources localized near the incident direction. Its proof is given in Appendix B.3.

Lemma 3.2. *Let $0 < \mu < 2$, $0 < a < R$, and $T > 0$ be fixed, and let $d \in \mathbb{S}^2$. Write $x = p + td$, where $p \in d^\perp$, and set*

$$K_d = \{p + td : |p| \le a,\ |t| \le T\}, \qquad \Gamma = \partial B_R.$$

There is a fixed $\varepsilon > 0$ with the following property. Let $q_d \in C_c^\infty$ be supported in $|\xi - d| < \varepsilon$, with uniformly bounded symbol seminorms under rotations taking d to a fixed axis. Then, for $0 < h \le 1$ and $\operatorname{supp} f \subset K_d$,

$$\|R_{h,0} q_d(hD) f\|_{L^2(\Gamma)} \le C h^{-1} \|f\|_2, \qquad R_{h,0} = (|hD|^\mu - 1 - i0)^{-1}. \tag{3.9}$$

The source may meet Γ. Moreover, if K_D and K_O are fixed bounded source and observation sets with

$$y\cdot d \ge T_+ \quad (y \in K_D), \qquad x\cdot d \le T_- \quad (x \in K_O), \qquad T_- < T_+,$$

then, for every fixed integer N,

$$\|R_{h,0} q_d(hD) f\|_{L^\infty(K_O)} \le C_N h^N \|f\|_2, \qquad \operatorname{supp} f \subset K_D. \tag{3.10}$$

All constants are uniform in d.

The total field $u_{V,h,d} = u_V(\cdot, d, h^{-1})$ admits the following approximation.

Lemma 3.3. *Let $0 < \mu < 2$ and $V \in \mathcal{V}^4_{r,F}$. For the phase and amplitude in Lemma 3.1, there are $h_0, C > 0$ such that the total field satisfies*

$$u_{V,h,d} = e^{iS_{\delta,V,d}/h}(a_{0,\delta,V,d} + \rho_{V,h,d}), \qquad \sup_{d\in\mathbb{S}^2} \|\rho_{V,h,d}\|_{L^2(B_R)} \le Ch^{1+\mu}, \quad 0 < h \le h_0. \tag{3.11}$$

Proof. Write $u = u_{V,h,d}$ and abbreviate the phase and amplitude of Lemma 3.1 by S and a_0. Extend $S - x\cdot d$ and $a_0 - 1$ by fixed cutoffs outside a larger construction slab, preserving them near the observation and cutoff sets. For $S^{\mathrm{ext}} = x\cdot d + \chi(S - x\cdot d)$, the formula

$$\nabla S^{\mathrm{ext}} = d + \chi(\Xi - d) + (\nabla\chi)(S - x\cdot d)$$

retains the fixed-parameter C^3 momentum bound. The corresponding extension $1 + \chi(a_0 - 1)$ retains the C^2 amplitude bound and positivity. We continue to write S, a_0 for these extensions.

The directional decomposition. Choose fixed numbers

$$r < r_0 < r_1 < r_2 < R.$$

With a sufficiently small fixed upper bound on δ, trajectories at transverse distance at least r_0 from the incident axis do not reach $\operatorname{supp} V$ in the construction slab, since their displacement from free trajectories is uniformly $O(\delta)$ on the fixed time interval. The extended function

$$W_h = e^{iS/h}a_0 - e^{ix\cdot d/h}$$

therefore has support in a fixed bounded tube with $|x - (x\cdot d)d| \le r_0$. The extension cutoffs preserve this support property.

Let $q_d \in C_c^\infty(\mathbb{R}^3)$ equal one in a neighborhood of d and have support in a sufficiently small fixed ball about d. Its support is disjoint from $|\xi| \le 1/4$. Decrease the admissible upper bound on δ so that $q_d = 1$ on a fixed neighborhood of every extended phase gradient. Put

$$Q_h = q_d(hD), \qquad Q_h^\perp = I - Q_h.$$

Choose this radius small relative to $r_2 - r_1$ and the longitudinal size of the construction region. All these choices are independent of h and of the particular potential.

Choose $\chi_* \in C_c^\infty$ equal to one near $\overline{B_R}$, with downstream derivatives in a compact subslab of $2R < x\cdot d < 3R$. Its incoming and transverse derivative sets have fixed neighborhoods where $S = x\cdot d$ and $a_0 = 1$. The phase and amplitude retain their eikonal and transport equations on a neighborhood of $\operatorname{supp}\chi_*$. Put $b_0(\xi) = |\xi|^\mu - p_{\mathrm{sm}}(\xi)$ and set

$$U = e^{iS/h}a_0, \qquad b_h = \chi_* W_h, \qquad u_{\mathrm{app}} = e^{ix\cdot d/h} + b_h.$$

For $P_0 = |hD|^\mu - 1$, the exact identities are

$$\begin{aligned} E = P_{h,V}u_{\mathrm{app}} &= P_0 b_h + \delta V u_{\mathrm{app}} \\ &= E_I + \mathcal{C}_h + b_0(hD)b_h, \\ E_I &= \chi_*(p_{\mathrm{sm}}(hD) - 1 + \delta V)U, \\ \mathcal{C}_h &= [p_{\mathrm{sm}}(hD), \chi_*]W_h. \end{aligned} \tag{3.12}$$

Apply the finite symbol expansion (B.3) from Appendix B.1 with $L = 2$. The C^2 amplitude bound suffices, and the eikonal and transport equations cancel the first two coefficients, giving

$$E_I = h^2\delta e^{iS/h}b_{I,h}, \qquad \|b_{I,h}\|_\infty \le C, \qquad \operatorname{supp} b_{I,h} \subset \operatorname{supp}\chi_*.$$

Choose a smooth transverse cutoff χ_T equal to one on a fixed neighborhood of the tube of radius r_0, and supported in the tube of radius r_1. Write $E_{I,T} = \chi_T E_I$. Then

$$\|E_{I,T}\|_2 \le Ch^2\delta, \qquad \operatorname{supp} E_{I,T} \subset \{y : |y - (y\cdot d)d| \le r_1,\ |y\cdot d| \le D\}, \tag{3.13}$$

where D is fixed. The remainder $(1-\chi_T)E_I$ has weighted L^2 and L^∞ norms $O(h^M)$ for every fixed M. On its support, W_h and V vanish locally, so the smooth-symbol kernel acts across a fixed positive transverse separation. Its rapid decay away from the diagonal gives arbitrary powers of h from the uniform L^1 bound of W_h.

Let $\chi_D \in C_c^\infty$ equal one near the downstream derivative support of χ_* and have support in $2R < x\cdot d < 3R$. Put $E_D = \chi_D \mathcal{C}_h$. The incoming and transverse parts of $\mathcal{C}_h$ have arbitrary-order tails because W_h vanishes on fixed neighborhoods there. Hence we have the exact decomposition

$$E = E_{I,T} + E_D + E_T + b_0(hD)b_h, \qquad \|\langle x\rangle E_T\|_2 + \|E_T\|_\infty \le C_M h^M \tag{3.14}$$

for every prescribed M. The commutator expansion (B.5) with $L = 2$ gives $\|E_D\|_1 + \|E_D\|_2 \le Ch$ without additional spatial derivatives.

Fix $\overline{B_R} \subset \Omega_c \Subset B_{3R/2}$. The downstream source is separated from this set in the d direction. Equation (3.10) gives, for every fixed M,

$$\|R_{h,0}Q_hE_D\|_{L^2(\Omega_c)} + \|R_{h,0}Q_hE_D\|_{L^\infty(\Gamma)} \le C_M h^M. \tag{3.15}$$

Since the source $E_{I,T}$ satisfies the tube condition of Lemma 3.2, (3.9) gives

$$\|R_{h,0}Q_hE_{I,T}\|_{L^2(\Gamma)} \le Ch^{-1}\|E_{I,T}\|_2 \le Ch^{1+\mu}. \tag{3.16}$$

The complementary frequencies. For a compactly supported C^2 amplitude a, the finite expansion of $Q_h^\perp(e^{iS/h}a)$ has vanishing coefficients of orders zero and one. Its kernel is a delta distribution minus a Schwartz kernel, and the remainder is bounded by

$$\left\|Q_h^\perp(e^{iS/h}a)\right\|_\infty + \left\|\langle x\rangle Q_h^\perp(e^{iS/h}a)\right\|_2 \le Ch^2\|a\|_{C^2}. \tag{3.17}$$

Outside a fixed neighborhood of the amplitude support, it has arbitrary-order Schwartz tails.

Put $z_h = Q_h^\perp b_h$ and $F_h = Q_h^\perp(Vu_{\mathrm{app}})$. For $\mu \le 1$, apply (3.17) separately to the two phases in b_h. For $\mu \ge 1$, write

$$b_h = e^{ix\cdot d/h}\chi_* A_h, \qquad A_h = e^{i(S-x\cdot d)/h}a_0 - 1, \qquad \|A_h\|_{C^2} \le C\delta/h.$$

Since $Vu_{\mathrm{app}} = Ve^{iS/h}a_0$, the second filtered input has a compactly supported amplitude with a uniform C^2 bound. Thus, with $\kappa_\mu = \max(2, 1+\mu)$,

$$\|z_h\|_\infty + \|\langle x\rangle z_h\|_2 \le Ch^{\kappa_\mu}, \qquad \|F_h\|_\infty + \|\langle x\rangle F_h\|_2 \le Ch^2. \tag{3.18}$$

For any fixed $r < r' < R$, the part of F_h outside $B_{r'}$ has arbitrary-order weighted L^2 and pointwise bounds, by the Schwartz kernel of $q_d(hD)$ and the support of Vu_{app} in B_r.

The exact resolvent identities give

$$\begin{aligned} R_{h,0}Q_h^\perp E &= z_h + \delta R_{h,0}F_h, \\ -R_{h,V}Q_h^\perp E &= -z_h + \delta R_{h,V}(Vz_h) - \delta R_{h,V}F_h. \end{aligned} \tag{3.19}$$

The second identity uses $R_{h,V}P_0 z_h = z_h - \delta R_{h,V}(Vz_h)$. Here $z_h \in H^2$ and is rapidly decreasing outside a fixed compact set. Since $0 < \mu < 2$, the function $P_0 z_h$ lies in weighted L^2; its

fractional tail is $O(|x|^{-3-\mu})$. The outgoing identities follow for smooth potentials and then by approximation. The weighted resolvent estimate gives

$$\left\|R_{h,V}Q_h^{\perp}E\right\|_{L^2(B_R)} \le C\Big(h^{\kappa_\mu} + h^{\kappa_\mu+\mu-1} + h^{1+\mu}\Big) \le Ch^{1+\mu}.$$

For the free sphere estimate, split F_h with a smooth cutoff supported in $B_{r'}$ and equal to one near B_r. The separated receiver estimate (B.7) controls the compact part by $Ch^{-1}h^2$. The weighted classical kernel bound and the fractional correction estimate (B.6) control the remaining arbitrary-order tail. Consequently

$$\left\|R_{h,0}Q_h^{\perp}E\right\|_{L^2(\Gamma)} \le C\Big(h^{\kappa_\mu} + h^{1+\mu}\Big). \tag{3.20}$$

Comparison with the outgoing field. For the near frequency part, $Q_h b_0(hD) = 0$ exactly. Apply the weighted perturbed resolvent to $Q_h E_{I,T}$ and the arbitrary-order tail. The former has weighted L^2 norm $Ch^2\delta$, since Q_h is uniformly bounded on that space. For the downstream term, use

$$R_{h,V}Q_hE_D = R_{h,0}Q_hE_D - \delta R_{h,V}\Big(VR_{h,0}Q_hE_D\Big)$$

and (3.15). The error is $u - u_{\mathrm{app}} = -R_{h,V}E$. These estimates, together with (3.19), give

$$\|u - u_{\mathrm{app}}\|_{L^2(B_R)} \le Ch^{1+\mu}. \tag{3.21}$$

Since $u_{\mathrm{app}} = e^{iS/h}a_0$ near $\overline{B_R}$, this proves (3.11).

For $V \in C_c^4(B_r)$, take a smooth approximation in C^4 supported in B_r. The flow, momentum, and incoming label converge jointly in C^3, and the leading amplitude converges in C^2. The two path identities used in (3.6) give convergence of $\partial_\delta^3 S$ in $C^1_{x,d}$. The finite kernel remainders converge in the bounded norms used above. The localized Fredholm inverse is continuous at the common admissible high frequencies. The constants depend only on the stated C^4 bound and fixed geometry, so the estimates pass to the limit. □

3.3. **Approximation on the receiver sphere.** The volume error also controls the feedback from the potential support to the receiver sphere.

Lemma 3.4. *Under the hypotheses of Lemma 3.3, the same phase and positive amplitude satisfy*

$$\sup_{d\in\mathbb{S}^2}\left\|u_{V,h,d} - e^{iS_{\delta,V,d}/h}a_{0,\delta,V,d}\right\|_{L^2(\Gamma)} \le Ch^{\nu_\mu}, \qquad \nu_\mu = \min(2\mu, 1+\mu), \quad 0<\mu<2. \tag{3.22}$$

The constants depend only on μ, r, R, F.

Proof. Use the approximate field u_{app} and residual E from the proof of Lemma 3.3. The integral equation gives

$$u - u_{\mathrm{app}} = -R_{h,0}E - \delta R_{h,0}\Big(V(u - u_{\mathrm{app}})\Big).$$

Use (3.16) on $E_{I,T}$, (3.15) on E_D, and (3.20) on the complementary part. The Schwartz kernel of Q_h preserves the weighted L^2 and L^∞ tail bounds uniformly in h. For the tails, Cauchy–Schwarz against the classical kernel and

$$\sup_{x\in\Gamma}\int_{\mathbb{R}^3}\frac{\mathrm{d}y}{\langle y\rangle^2|x-y|^2} < \infty$$

give a factor Ch^{-2} from the weighted L^2 norm. The fractional correction is bounded by (B.6) and the L^∞ norm. Choosing $M \geq 5$ in (3.14) bounds these terms by Ch^3. The separated receiver estimate (B.7) and the volume error (3.21) control the feedback term by $Ch^{2\mu}$. It follows that

$$\left\|u - e^{iS/h}a_0\right\|_{L^2(\Gamma)} \leq C\Big(h^{1+\mu} + h^{\kappa_\mu} + h^{2\mu} + h^3\Big) \leq Ch^{\min(2\mu,1+\mu)}. \qquad \square$$

For $\mu \leq 1$, the squared spherical error is $O(h^{4\mu})$, the order used in the real-frequency estimate below.

4. Increasing stability for the inverse scattering problem

In this section, we prove Theorem 1.1 which establishes the increasing stability for the inverse potential scattering problem. Let $W = V_1 - V_2$ and retain $\mu = 2\alpha$. We use ray transforms and integer phase combinations to obtain a real-frequency estimate, then apply analytic continuation to choose the frequency window.

4.1. **Ray estimates and phase differences.**

Lemma 4.1. *For every $f \in L^2(\mathbb{R}^3)$ supported in B_r and either observation set in* (1.4),

$$\|f\|_{H^{-1/2}(\mathbb{R}^3)} \leq C\,\|Bf\|_Y\,. \tag{4.1}$$

Proof. Parameterize oriented lines by (p, d), where $p \in d^\perp$. The Fourier slice identity and Plancherel's theorem give

$$\int_{\mathbb{S}^2}\int_{d^\perp} |Xf(p,d)|^2\,\mathrm{d}p\,\mathrm{d}d = c\int_{\mathbb{R}^3} |\xi|^{-1}|\widehat{f}(\xi)|^2\,\mathrm{d}\xi \geq c'\,\|f\|^2_{H^{-1/2}}\,. \tag{4.2}$$

Compact support and L^2 regularity give $f \in L^1$, so the integral is finite near the origin.

For spherical observations, parameterize the downstream hemisphere by $x = p + \sqrt{R^2 - |p|^2}\,d$. The half-ray integral equals $Xf(p,d)$ at this point and vanishes at the upstream intersection. Therefore,

$$\int_\Gamma |Bf(x,d)|^2\,\mathrm{d}\sigma(x) = \int_{|p|<r} \frac{R}{\sqrt{R^2-|p|^2}}|Xf(p,d)|^2\,\mathrm{d}p. \tag{4.3}$$

The Jacobian is bounded above and below on $|p| < r < R$. Combining this identity with (4.2) proves (4.1) for $\mathcal{O} = \Gamma$.

For annular observations, the part of a line in A has length

$$\ell(p) = 2\Big(\sqrt{R^2-|p|^2} - \sqrt{r^2-|p|^2}\Big), \qquad 2(R-r) \leq \ell(p) \leq 2R$$

when $|p| < r$. Since Xf is constant on each line and vanishes for $|p| \geq r$, Fubini's theorem gives

$$\|Xf\|^2_{L^2(A\times\mathbb{S}^2)} = \int_{\mathbb{S}^2}\int_{d^\perp} \ell(p)|Xf(p,d)|^2\,\mathrm{d}p\,\mathrm{d}d.$$

The result follows from (4.2) and $Xf = Bf(\cdot,d) + Bf(\cdot,-d)$. $\square$

We next compare the phases associated with the two potentials. Set $G(k) = u_1^s(k) - u_2^s(k)$, $\beta = BW$, and

$$\phi_j(x, d, k) = k(S_{k^{-\mu}, V_j, d}(x) - x \cdot d), \qquad \psi = \phi_1 - \phi_2.$$

The joint regularity in the independent parameter δ yields the following phase expansion.

Lemma 4.2. *There is a real function $c_2(x, d)$, independent of k, such that, for sufficiently large k,*

$$\begin{aligned} \psi(k) &= -\frac{\beta}{\mu} k^{1-\mu} + c_2 k^{1-2\mu} + \mathfrak{r}(k), \\ \|\mathfrak{r}(k)\|_{C^1} \le Ck^{1-3\mu}, &\qquad \|\partial_k \mathfrak{r}(k)\|_{C^1} \le Ck^{-3\mu}. \end{aligned} \tag{4.4}$$

Here the C^1 norm is taken in the spatial and incident-direction variables on the fixed construction sets.

Proof. The mixed bound (3.6) in the phase construction gives a uniformly bounded third δ derivative of the phase in $C^1_{x,d}$. For $D_\delta = S_{\delta,V_1,d} - S_{\delta,V_2,d}$, Taylor's formula therefore gives

$$D_\delta = -\delta\beta/\mu + \delta^2 c_2 + r_\delta, \qquad c_2 = \tfrac{1}{2}\partial_\delta^2 D_0, \qquad r_\delta = \tfrac{1}{2}\int_0^\delta (\delta - t)^2 \partial_\delta^3 D_t \, \mathrm{d}t.$$

Thus $\|r_\delta\|_{C^1} \le C\delta^3$ and $\|\partial_\delta r_\delta\|_{C^1} \le C\delta^2$. Taking $\mathfrak{r}(k) = k r_{k^{-\mu}}$ and using $\partial_k \mathfrak{r} = r_\delta - \mu\delta\partial_\delta r_\delta$ proves the assertion. □

Positive leading amplitudes yield the following estimate for integer combinations of phases.

Lemma 4.3. *Let $t_1, \dots, t_m > 0$ and $n_1, \dots, n_m \in \mathbb{Z}$ be fixed, and set $\Psi(k) = \sum_{\ell=1}^m n_\ell \psi(t_\ell k)$. Put $\nu_\mu = \min(2\mu, 1 + \mu)$. Then, uniformly in d and for sufficiently large k,*

$$\left\| e^{i\Psi(k)} - 1 \right\|^2_{L^2(\mathcal{O})} \le C \left(\sum_{\ell=1}^m \|G(\cdot, d, t_\ell k)\|^2_{L^2(\mathcal{O})} + k^{-2\nu_\mu} \right). \tag{4.5}$$

The same inequality holds in Y after integration in d. The constant and the frequency threshold may also depend on the fixed ratios t_ℓ and integers n_ℓ.

Proof. Lemmas 3.3 and 3.4 give $e^{-ikx\cdot d} u_j = e^{i\phi_j}(a_j + \rho_j)$, with $a_j \ge 3/4$ real and $\|\rho_j(\cdot, d, k)\|_{L^2(\mathcal{O})} \le Ck^{-\nu_\mu}$. The identity

$$|a_1 e^{i\phi_1} - a_2 e^{i\phi_2}|^2 = (a_1 - a_2)^2 + a_1 a_2 |e^{i\psi} - 1|^2$$

implies $|e^{i\psi(k)} - 1|^2 \le C(|G(x, d, k)|^2 + |\rho_1|^2 + |\rho_2|^2)$ almost everywhere. Telescoping on the unit circle gives

$$|e^{i\Psi(k)} - 1| \le \sum_{\ell=1}^m |n_\ell| |e^{i\psi(t_\ell k)} - 1|.$$

Squaring and integrating proves (4.5). □

4.2. **The real-frequency estimate.** Integer combinations of phases remove the second coefficient in (4.4). For $\mu < 1$, frequency averaging controls the resulting oscillations; at $\mu = 1$, the incoming cap fixes the phase branch; for $\mu > 1$, the phase tends to zero. These arguments give a uniform estimate on a fixed relative window.

Proposition 4.4. *Set $\tau = (\mu - 1)_+$ and $\rho = \min(4\mu, 4)$. There are $K_*, C > 0$, depending only on α, r, R, F, such that, for every $Q \geq 4K_*$,*

$$\|W\|^2_{H^{-1/2}(\mathbb{R}^3)} \leq C \left(\frac{1}{Q} \int_{Q/4}^{Q} t^{2\tau} \, \|G(t)\|_Y^2 \, \mathrm{d}t + Q^{-\rho} \right). \tag{4.6}$$

Proof. By Lemma 4.1, it suffices to estimate $\|\beta\|_Y^2$. We treat the three ranges of μ separately.

The case $0 < \mu < 1$. Take $a = 3/4$, $p = 1 - 2\mu$, and

$$\lambda = \begin{cases} (2 - a^p)^{1/p}, & p \neq 0, \\ 4/3, & p = 0. \end{cases}$$

The power mean of order p of a, λ equals one. Comparison with the orders -1 and 1 gives $5/4 \leq \lambda \leq 3/2$. Since $1 - \mu > p$, strict monotonicity of power means gives $c_\mu = a^{1-\mu} + \lambda^{1-\mu} - 2 > 0$. Thus $\Psi(k) = \psi(ak) - 2\psi(k) + \psi(\lambda k)$ cancels the second coefficient in (4.4), and

$$\Psi'(k) = -\frac{c_\mu(1-\mu)}{\mu} \beta k^{-\mu} + O(k^{-3\mu}). \tag{4.7}$$

Fix (x, d) and put $I_Q = [Q/(2\lambda), Q/\lambda]$. We have $|\beta(x, d)| \leq M_B$ uniformly. If $|\beta| \leq A_0 Q^{-2\mu}$, its square is bounded by $A_0^2 Q^{-4\mu}$. Otherwise choose the fixed A_0 large enough that (4.7) yields

$$c|\beta|Q^{-\mu} \leq |\Psi'(k)| \leq C|\beta|Q^{-\mu} \quad (k \in I_Q),$$

with a fixed sign. The phase is therefore monotone on I_Q, with image length comparable to $T = Q^{1-\mu}|\beta|$. For every real interval J of length ℓ,

$$\frac{1}{\ell} \int_J |e^{is} - 1|^2 \, \mathrm{d}s \geq c \min(1, \ell^2);$$

indeed, the minimum over its location is $2 - 4|\sin(\ell/2)|/\ell$. Changing variables to $s = \Psi(k)$ gives

$$\frac{1}{Q} \int_{I_Q} |e^{i\Psi(k)} - 1|^2 \, \mathrm{d}k \geq c \min(1, T^2) \geq c_{M_B} |\beta|^2.$$

Including the small-β case, integrating in (x, d), and applying Lemma 4.3, we obtain

$$\|\beta\|_Y^2 \leq C \left(\frac{1}{Q} \int_{Q/4}^{Q} \|G(t)\|_Y^2 \, \mathrm{d}t + Q^{-4\mu} \right).$$

Here $aI_Q, I_Q, \lambda I_Q \subset [Q/4, Q]$, since $a/(2\lambda) \geq 1/4$. Since $2\nu_\mu = 4\mu$, the field approximation contributes the same remainder order.

The case $\mu = 1$. Set $\Psi(k) = 2\psi(2k) - \psi(k)$. Lemma 4.2 gives

$$\Psi(k) = -\beta + O_{C^1}(k^{-2}). \tag{4.8}$$

Fix d, put $Z = e^{i\Psi(k)} - 1$, and write

$$e_d^2 = \|G(\cdot, d, k)\|^2_{L^2(\mathcal{O})} + \|G(\cdot, d, 2k)\|^2_{L^2(\mathcal{O})}.$$

The spatial Lipschitz norm of Z is uniformly bounded, and Lemma 4.3 gives $\|Z\|_{L^2(\mathcal{O})} \leq C(e_d + k^{-2})$. The observation measure satisfies $m_{\mathcal{O}}(\mathcal{O} \cap B(x,t)) \geq ct^{n_{\mathcal{O}}}$ for small t, where $n_{\mathcal{O}} = 2$ on the sphere and 3 on the annulus. Integrating on a ball whose radius is proportional to the maximum of $|Z|$ therefore yields

$$\|Z\|_{\infty} \leq C(e_d + k^{-2})^{2/(n_{\mathcal{O}}+2)}.$$

Choose a fixed small $e_* > 0$ and then increase K_*. If $e_d \leq e_*$, the last bound makes $\|Z\|_{\infty} < 1$. Since the observation set is connected and $\Psi = 0$ on its nonempty incoming cap, the phase lies in the component containing zero of $\{s : |e^{is} - 1| < 1\}$. Hence $|\Psi| \leq C|Z|$ on that component. Equation (4.8) gives

$$\|\beta(\cdot, d)\|^2_{L^2(\mathcal{O})} \leq C(e_d^2 + k^{-4}).$$

For the remaining directions, $e_d > e_*$, the uniform a priori bound for β gives the same inequality. Integrating in d and then averaging over $k \in [Q/4, Q/2]$ proves

$$\|\beta\|^2_Y \leq C\left(\frac{1}{Q}\int_{Q/4}^{Q} \|G(t)\|^2_Y \, \mathrm{d}t + Q^{-4}\right).$$

The case $1 < \mu < 2$. Take $\lambda = 2^{1/(2\mu-1)} \in (1,2)$ and $\Psi(k) = 2\psi(\lambda k) - \psi(k)$. Since $2\lambda^{1-2\mu} = 1$, we have

$$\Psi(k) = -\frac{d_\mu \beta}{\mu} k^{1-\mu} + O(k^{1-3\mu}),$$
$$d_\mu = 2\lambda^{1-\mu} - 1 = 2^{\mu/(2\mu-1)} - 1 > 0.$$

Since $\|\Psi\|_{\infty} \leq Ck^{1-\mu} \to 0$, we have $|\Psi| \leq C|e^{i\Psi} - 1|$ for sufficiently large k. Lemma 4.3 consequently gives

$$\|\beta\|_Y \leq C\left(k^{\mu-1}\Big(\|G(k)\|_Y + \|G(\lambda k)\|_Y\Big) + k^{-2} + k^{-2\mu}\right).$$

The last term is bounded by k^{-2}. Squaring and averaging over $k \in [Q/(2\lambda), Q/\lambda]$, for which $k, \lambda k \in [Q/4, Q]$, gives

$$\|\beta\|^2_Y \leq C\left(\frac{1}{Q}\int_{Q/4}^{Q} t^{2(\mu-1)} \|G(t)\|^2_Y \, \mathrm{d}t + Q^{-4}\right).$$

Finally, Lemma 4.1 proves (4.6) in all three cases. Choose K_* large enough that every sampled frequency satisfies the geometric optics and phase thresholds. □

4.3. **Analytic continuation of the measured fields.** We extend the measured fields from the fixed strip to a domain where quantitative continuation controls the data beyond the measured interval.

Proposition 4.5. *There are fixed $b > 0$ and $L > b + 1$, uniform for $V \in \mathcal{V}^4_{r,F}$, such that the localized outgoing resolvent and the Y-valued scattered field extend holomorphically to*

$$\mathcal{D}_{b,L} = \{z \in \mathbb{C} : \Im z > -b, \ |z + ib| > L\}. \tag{4.9}$$

Use the branch of $\log z$ *with cut along* $-i[0,\infty)$ *and argument in* $(-\pi/2, 3\pi/2)$. *For fixed compactly supported smooth cutoffs* η_1, η_2,

$$\|\eta_1 \mathcal{R}_{\alpha,V}(z)\eta_2\|_{L^2 \to L^2} \leq C|z|^{1-\mu}, \qquad z \in \mathcal{D}_{b,L}. \tag{4.10}$$

Moreover,

$$\|u_V^s(z)\|_Y \le C(1+|z|)^3 e^{r(\Im z)_+}. \tag{4.11}$$

These extensions agree with the outgoing fields on the sufficiently large positive real axis.

The spectral and outgoing resolvents agree on their first-quadrant overlap, and a spectral Sobolev estimate provides the spherical trace. The proof is given in Appendix A.6.

Corollary 4.6. *Let $G(z) = u_{V_1}^s(z) - u_{V_2}^s(z)$ and $\tau = (\mu-1)_+$. Fix $\kappa > b+1$ and an integer $N \ge 4$. Then*

$$F_N(z) = e^{irz}(z+i\kappa)^{-N} z^\tau G(z) \tag{4.12}$$

is holomorphic and uniformly bounded in Y on $\mathcal{D}_{b,L}$. Its norm on a measured positive real interval is at most the weighted discrepancy there, and

$$\|q^\tau G(q)\|_Y \le C(1+q)^N \|F_N(q)\|_Y \tag{4.13}$$

for positive real q in the domain.

Proof. The branch defining z^τ is holomorphic in the domain, and the pole $-i\kappa$ lies below it. Moreover,

$$|z+i\kappa| \ge \kappa - b > 1, \qquad 1+|z| \le C_{\kappa,b}|z+i\kappa|.$$

The factor $|e^{irz}| = e^{-r\Im z}$ cancels the exponential growth in (4.11) and is bounded below the real axis. Since $\tau < 1$ and $N \ge 4$, this proves the uniform bound. On the positive real axis the exponential has modulus one and $|q+i\kappa|^{-N} \le 1$, which proves the remaining assertions. □

4.4. **Choice of the frequency window.** In what follows, we apply the analytic continuation given by Lemma C.1 to the normalized field and balance the continued data against the real-frequency remainder, which yields the proof of Theorem 1.1.

Proof of Theorem 1.1. Put $\mu = 2\alpha$, $\rho = \min(4\mu, 4)$, $G(z) = u_1^s(z) - u_2^s(z)$, and $H(z) = z^\tau G(z)$. Fix $N = 4$ in Corollary 4.6 and choose $M \ge 1$, depending only on the a priori parameters, so that

$$F(z) = M^{-1} e^{irz}(z+i\kappa)^{-N} H(z)$$

is bounded by one in Y on $\mathcal{D}_{b,L}$. Choose $K_0 \ge 2L+2$ sufficiently large for all real-frequency estimates. On the measured interval, $\|F(k)\|_Y \le \varepsilon_w$. Lemma C.1 and (4.13) therefore yield

$$\|H(q)\|_Y \le C(1+q)^N \exp\left(-\frac{cE}{(1+q)^2}\right), \quad q \ge K_0+1, \qquad E = |\log \varepsilon_w|. \tag{4.14}$$

Set

$$\mathcal{L}(E) = \sqrt{\frac{E}{\log(2+E)}}.$$

Choose $0 < \sigma \le 1$ so small that

$$\frac{c}{4\sigma^2} > \frac{N}{2} + \frac{\rho}{4} + 1, \qquad a = \frac{\sigma}{8}. \tag{4.15}$$

If $K \ge a\mathcal{L}(E)$ and $K \ge 4K_0$, apply Proposition 4.4 with $Q = K$. Every frequency in $[K/4, K]$ is measured, and hence

$$\|W\|_{H^{-1/2}}^2 \le C(\varepsilon_w^2 + K^{-\rho}) \le C\left(\varepsilon_w^2 + \Big(K + \mathcal{L}(E)\Big)^{-\rho}\right).$$

If $K \geq a\mathcal{L}(E)$ but $K < 4K_0$, then $K + \mathcal{L}(E) \leq 4K_0(1 + a^{-1})$. The a priori bound gives the same estimate.

Suppose finally that $K < a\mathcal{L}(E)$ and take $Q = \sigma\mathcal{L}(E)$. Then

$$Q/4 = 2a\mathcal{L}(E) > 2K \geq 2(K_0 + 1),$$

so $[Q/4, Q]$ lies beyond the measured band and above the fixed frequency thresholds. For each k in this interval, (4.14) gives

$$\|H(k)\|_Y \leq CQ^N \exp\left(-\frac{cE}{4Q^2}\right) = CQ^N(2+E)^{-c/(4\sigma^2)} \leq CQ^{-\rho/2}.$$

The last inequality follows from $Q \leq E^{1/2}$ and (4.15). Substituting this bound into Proposition 4.4 yields

$$\|W\|^2_{H^{-1/2}} \leq CQ^{-\rho} \leq C\Big(K + \mathcal{L}(E)\Big)^{-\rho},$$

because $Q \asymp K + \mathcal{L}(E)$ in the present case. This proves (1.6).

For $0 < \theta < 1/2$, the ratio $\mathcal{L}(E)/E^\theta$ has a positive infimum on $[1, \infty)$ and tends to infinity as $E \to \infty$. Consequently $K + \mathcal{L}(E) \geq c_\theta(K + E^\theta)$, which proves the alternative form. The a priori H^s bound for $0 < s \leq 4$ and interpolation give

$$\|W\|^2_{L^2} \leq \|W\|^{4s/(2s+1)}_{H^{-1/2}} \|W\|^{2/(2s+1)}_{H^s}.$$

This proves the L^2 conclusion. The proof is complete. □

5. Numerical experiments

In this section, we present numerical examples to examine the effect of the measured frequency band on the reconstruction. For ease of visualization, we carry out the experiments in two dimensions.

We consider the two-dimensional fractional Helmholtz equation,

$$(-\Delta)^\alpha u - k^{2\alpha}u + Vu = 0 \quad \text{in } \mathbb{R}^2, \qquad u = e^{ikx\cdot d} + u^s, \quad d \in \mathbb{S}^1. \tag{5.1}$$

The complex scattered field is measured on a circle enclosing the support of V. The examples cover the three regimes $\alpha < 1/2$, $\alpha = 1/2$, and $\alpha > 1/2$.

5.1. **Reconstruction method.** Let $D = (-1, 1)^2$ and let $G^{(2)}_{\alpha,k}$ be the outgoing Green function of $(-\Delta)^\alpha - k^{2\alpha}$ in $\mathbb{R}^2$. Writing $u(x, d, k) = e^{ikx\cdot d}w(x, d, k)$ gives

$$w(x) + \int_D G^{(2)}_{\alpha,k}(x - y)e^{-ikd\cdot(x-y)}V(y)w(y)\,\mathrm{d}y = 1. \tag{5.2}$$

Factoring out the incident plane wave from the total field removes the known incident phase while retaining the full scattering interaction. On a square grid, let $T_{j\ell}$ and $H_{j\ell}$ be the cell integrals of the kernel in (5.2), evaluated at interior points and receivers, respectively. For $c \in \mathbb{R}^{81}$ and $V_c = \sum_{a=1}^{81} c_a\psi_a$, set $v = \Phi c$, where $\Phi_{na} = \psi_a(x_n)$. The discrete forward map is

$$A_{j\ell} = I + T_{j\ell}\,\mathrm{diag}(v), \qquad A_{j\ell}w_{j\ell} = \mathbf{1}, \qquad f_{j\ell}(c) = -H_{j\ell}(v \odot w_{j\ell}). \tag{5.3}$$

Here $\odot$ denotes entrywise multiplication. The entries of $f_{j\ell}$ are the scattered field multiplied by $e^{-ik_j z_m\cdot d_\ell}$; the measured data undergo the same known phase change.

Let $\widetilde{F}_j(c)$ collect these columns and let $\widetilde{Y}_j^{\eta}$ denote the noisy data. At upper endpoint K, we minimize

$$J_K(c) = \frac{\sum_{k_j \le K} \omega_{j,K} \|\widetilde{F}_j(c) - \widetilde{Y}_j^{\eta}\|_F^2}{2(K-2)N_r N_d\, s_\alpha^2} + \frac{\lambda_K}{2}|c|^2, \qquad \lambda_K = \frac{3.6 \times 10^{-4}}{K-2}. \tag{5.4}$$

The subscript F denotes the Frobenius norm, $N_r = 72$, $N_d = 36$, and $\omega_{j,K}$ are trapezoidal weights with sum $K-2$. The scale s_α is the root-mean-square value of the noisy $k=2$ data for V_1, fixed for all examples at that order. The penalty in the unnormalized cumulative functional is therefore independent of K.

The gradient is evaluated by an adjoint solve. For $r_{j\ell} = f_{j\ell} - \widetilde{Y}_{j,\ell}^{\eta}$ and $\gamma_{j,K} = \omega_{j,K}/((K-2)N_r N_d s_\alpha^2)$, set $b_{j\ell} = \gamma_{j,K} H_{j\ell}^* r_{j\ell}$ and solve $A_{j\ell}^* p_{j\ell} = \mathrm{diag}(v) b_{j\ell}$. Then

$$\nabla J_K(c) = \lambda_K c - \sum_{k_j \le K} \sum_{\ell=1}^{N_d} \Phi^T \,\mathrm{Re}\Big[\overline{w_{j\ell}} \odot (b_{j\ell} - T_{j\ell}^* p_{j\ell})\Big]. \tag{5.5}$$

We use BFGS with Armijo backtracking in coordinates $\zeta = L_K^T c$, where $L_K L_K^T$ is the regularized Born normal matrix. The states and gradients remain nonlinear. The initial vector is zero at $K=5$; each subsequent band starts from the preceding reconstruction. Iterations stop when the scaled gradient is below 10^{-6}, or when the objective decrease is below 2×10^{-11} together with a small step or gradient. At most three passes of 160 iterations are allowed.

5.2. **Experimental setting.** The receivers are $z_m = 2.2(\cos\theta_m, \sin\theta_m)$, where $\theta_m = 2\pi(m+1/2)/72$, $m = 0,\dots,71$. The incident angles are $2\pi\ell/36$, $\ell = 0,\dots,35$. We sample the wavenumbers $2,3,4,5,6,10,15,20,30,40,50$ and use the cumulative endpoints $K = 5,10,20,30,40,50$. At every sampled wavenumber, synthetic data are generated on 192^2 cells and inversion uses 128^2 cells. Cell integrals include the oscillatory factor in (5.2); singular self-interactions are integrated by a Duffy transformation. The interior operator is applied by zero-padded FFTs and the linear systems are solved by GMRES. Its relative tolerance is 2×10^{-8} for data generation and inversion, except that inversion at $\alpha = 0.25$ uses 2×10^{-10}.

For the reconstruction space, we take

$$\psi_a(x) = \chi(x)\exp\left(-\frac{|x-\xi_a|^2}{2(0.16)^2}\right), \qquad \xi_a \in \{-0.64,-0.48,\dots,0.64\}^2.$$

The radial cutoff is $\chi(x) = s((0.94-|x|)/0.16)$, where $s(t) = \rho(t)/(\rho(t)+\rho(1-t))$ and $\rho(t) = e^{-1/t}$ for $t>0$, with $\rho(t)=0$ otherwise. Thus $\chi = 1$ on $B_{0.78}$ and vanishes outside $B_{0.94}$. To define the test potentials, put

$$g(x;a,b,\vartheta) = \exp\left(-\frac{1}{2}\sum_{i=1}^{2}\frac{[R_\vartheta^T(x-a)]_i^2}{b_i^2}\right),$$

where R_ϑ is the rotation matrix. We choose

$$\begin{aligned}
V_1(x) &= \chi(x) g(x; (0.12, -0.08), (0.24, 0.34), \pi/7),\\
V_2(x) &= \chi(x)\Big[g(x; (-0.35, -0.18), (0.22, 0.22), 0)\\
&\qquad + 0.8 g(x; (0.35, 0.20), (0.24, 0.24), 0)\Big],\\
V_3(x) &= \chi(x)\Big[1.1 g(x; (-0.28, 0.20), (0.28, 0.14), \pi/6)\\
&\qquad - 0.85 g(x; (0.32, -0.20), (0.20, 0.25), -\pi/5)\Big].
\end{aligned}$$

These are a positive peak, two positive peaks, and a sign-changing pair. All three are smooth and compactly supported.

At each frequency, we perturb the clean data Y_j by

$$Y_j^\eta = Y_j + \eta \|Y_j\|_F \frac{E_j}{\|E_j\|_F}, \qquad \eta = 0.03, \tag{5.6}$$

where the real and imaginary parts of E_j are independent standard Gaussian arrays. One fixed realization is used for each order, potential and frequency, and is retained when the band expands. All reconstruction errors are evaluated on a common 161^2 grid using $e_V(K) = \|V_{c_K} - V\|_{L^2(D)}/\|V\|_{L^2(D)}$. The exact potentials enter data generation and error evaluation, but not parameter or iterate selection.

5.3. **Reconstruction results.** Table 1 gives all relative reconstruction errors. Figures 1–3 show the reconstructions at all six endpoints, together with their absolute errors. The final errors range from 0.2448% to 1.2564%; the corresponding data residuals are 2.997%–3.046%, close to the prescribed noise level.

TABLE 1. Relative L^2 reconstruction errors (%) with 3% noise.

α	Potential	$K = 5$	10	20	30	40	50
0.25	V_1	1.6153	0.6041	0.3128	0.2584	0.2504	0.2448
	V_2	2.8857	0.4750	0.3116	0.2910	0.2829	0.2709
	V_3	7.6540	1.0696	0.6988	0.6616	0.6543	0.6489
0.50	V_1	0.8385	0.4980	0.3729	0.3578	0.3679	0.3700
	V_2	1.2799	0.5240	0.4365	0.4189	0.4023	0.3870
	V_3	8.0690	1.2532	0.8587	0.8054	0.7724	0.7420
0.75	V_1	0.8639	0.5202	0.4087	0.3715	0.3608	0.3534
	V_2	1.3511	0.4903	0.4336	0.3932	0.3790	0.3661
	V_3	9.1701	2.6255	1.5397	1.3840	1.3054	1.2564

All nine examples improve between $K = 5$ and $K = 50$; eight curves decrease monotonically. The exception is V_1 at $\alpha = 0.5$, whose error rises from 0.3578% at $K = 30$ to 0.3700% at $K = 50$. The computed projection errors on the evaluation grid are 0.1833%, 0.2053%, and 0.6198% for V_1, V_2, V_3. The final V_3 error 0.6489% at $\alpha = 0.25$ is close to this level. Figure 4 collects these error curves and projection levels.

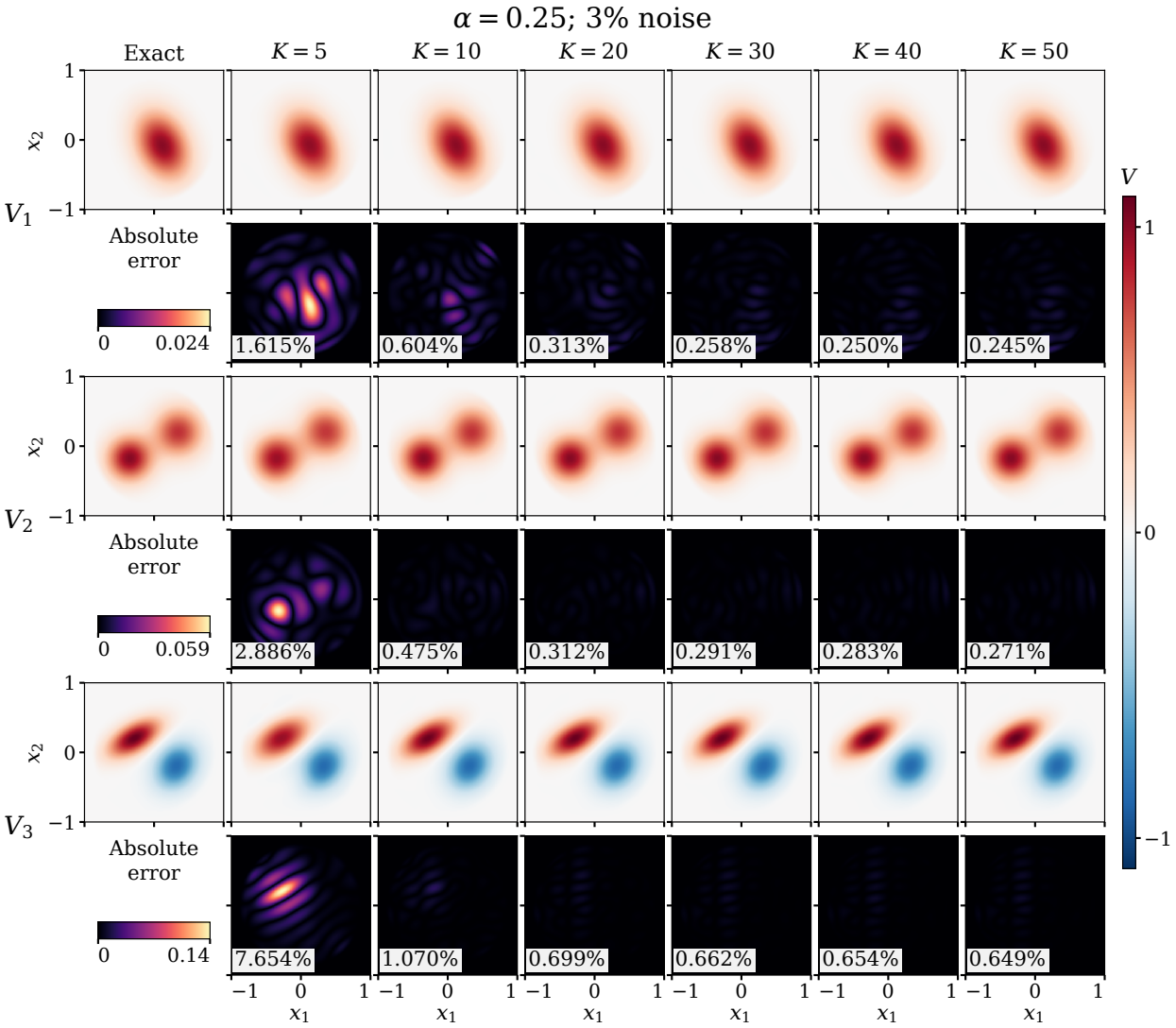


FIGURE 1. Reconstructions and absolute errors for $\alpha = 0.25$. Row pairs correspond to V_1, V_2, V_3, with reconstruction above error. The first column contains exact potentials and error colorbars; the remaining columns show $K = 5, 10, 20, 30, 40, 50$. The right colorbar applies to potentials. For each potential, the error scale is fixed across all K and α. Labels give relative L^2 errors (%).

Table 2 compares nonlinear and Born reconstructions of V_3 at $K = 50$. The Born fits set $w_{j\ell} = \mathbf{1}$ in (5.3), with the same data, basis, weights and penalty. The improvement is largest for $\alpha = 0.25$.

TABLE 2. Born and nonlinear reconstruction errors (%) for V_3 at $K = 50$.

α	Born	Nonlinear
0.25	104.696	0.649
0.50	6.472	0.742
0.75	2.868	1.256

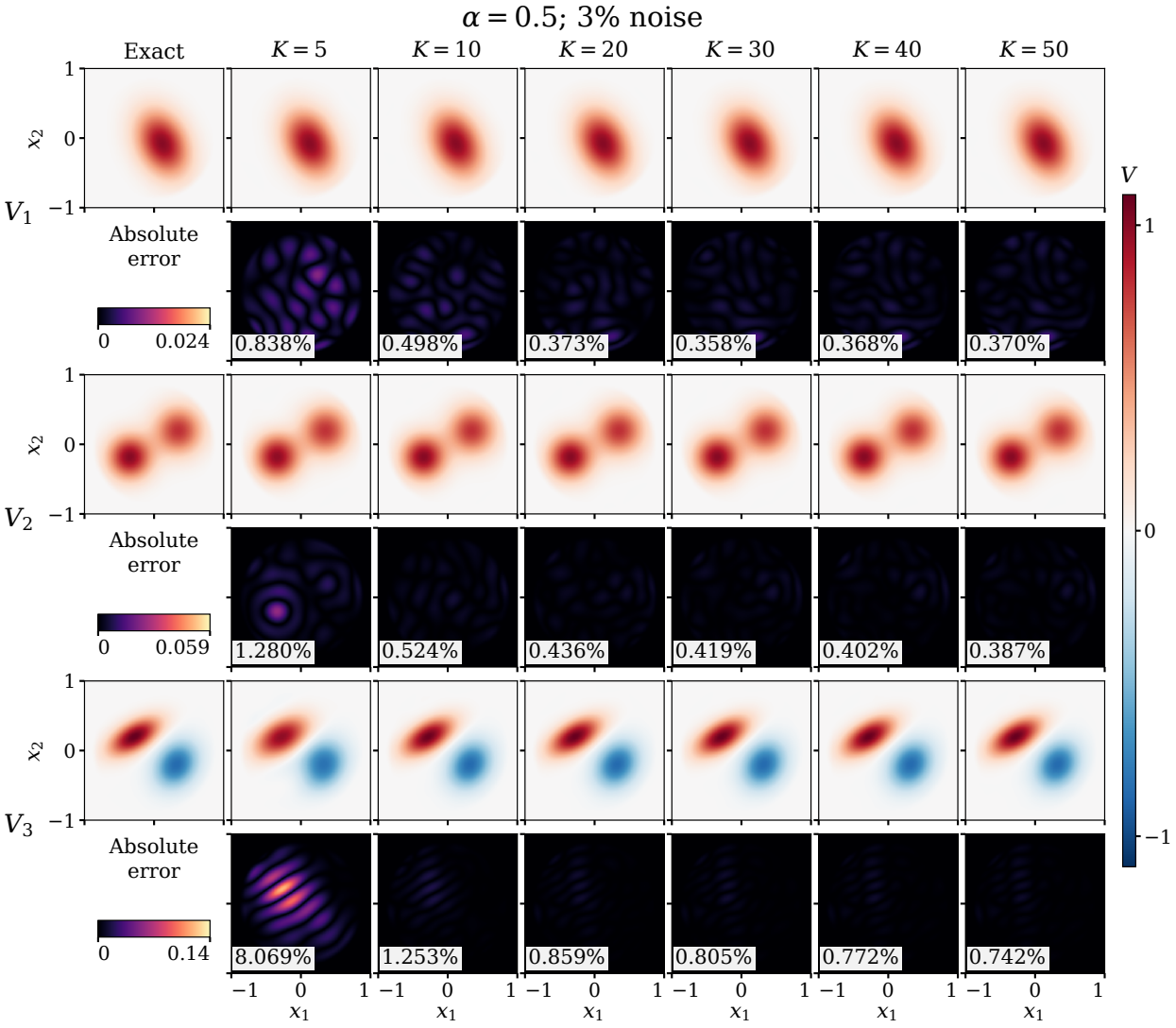


FIGURE 2. Reconstructions and absolute errors for $\alpha = 0.5$, with the layout and color scales of Figure 1.

The planar examples illustrate the bandwidth effect, without assessing the three-dimensional rate at fixed weighted absolute data discrepancy.

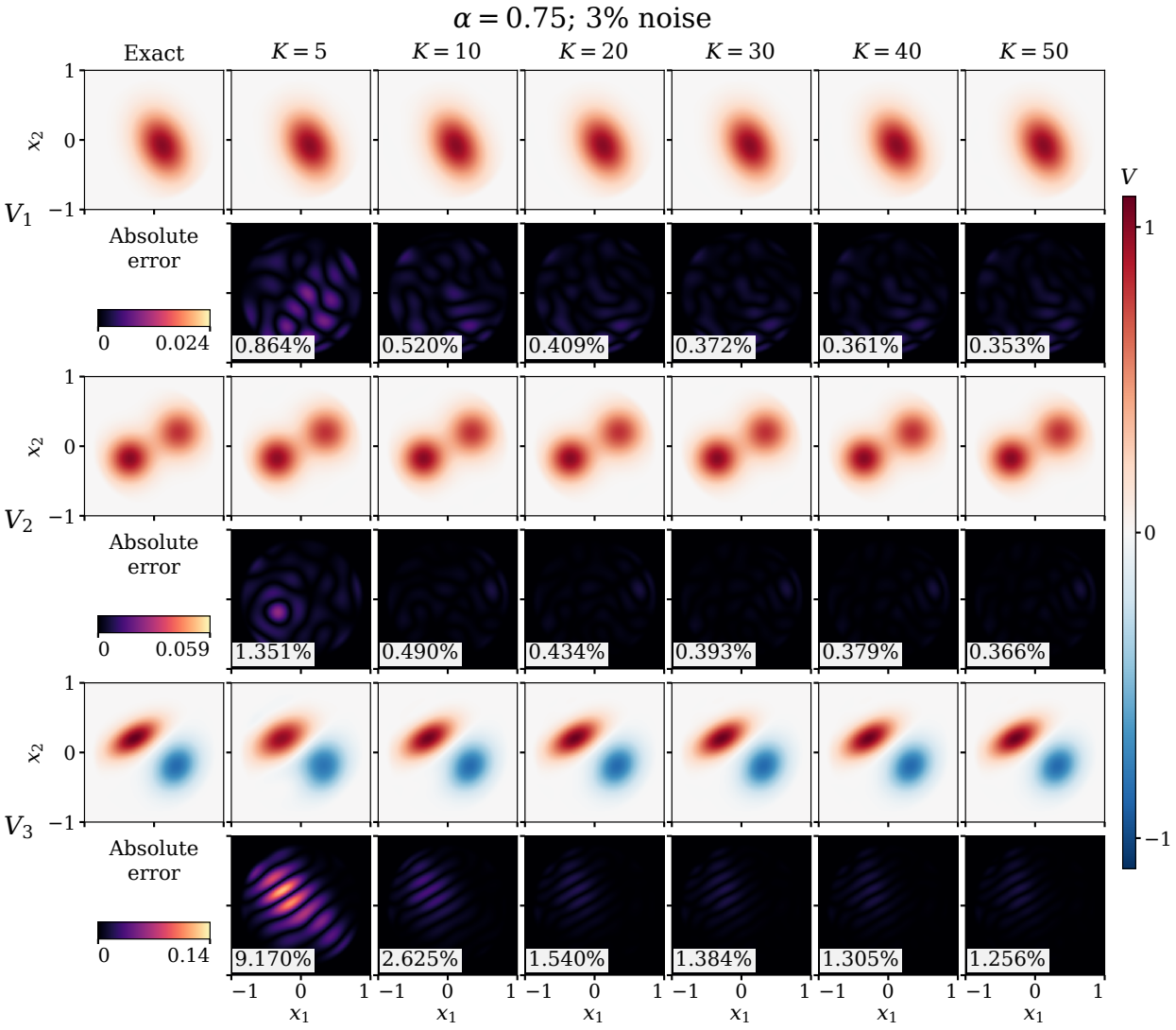


FIGURE 3. Reconstructions and absolute errors for $\alpha = 0.75$, with the layout and color scales of Figure 1.

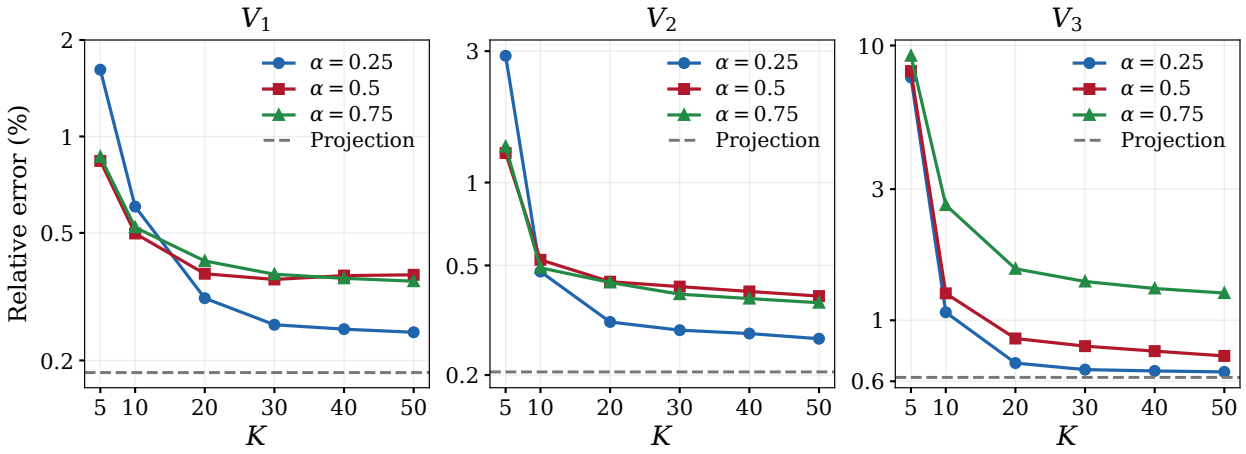


FIGURE 4. Relative errors versus K on logarithmic vertical axes. Dashed lines are the computed least-squares projection errors in the 81-dimensional space on the 161^2 evaluation grid.

6. Conclusion

For every $0 < \alpha < 1$, we have established high-frequency well-posedness of the fractional Helmholtz equation for real compactly supported C^1 potentials. Under a uniform C^4 bound, we have derived increasing stability from complex L^2 multifrequency measurements on an enclosing sphere or in an exterior annulus, for all incident directions. The proof relates the scattered fields to ray integrals through geometric optics, controls the nonlinear phase by frequency combinations and averaging, and extends the frequency range by analytic continuation.

Corresponding estimates under weaker regularity assumptions, with fewer incident directions, or from phaseless measurements remain open. We hope to report further progress on these problems in future work.

Appendix A. Resolvent and analytic continuation estimates

This appendix proves results on the direct scattering in Sections 2 and Proposition 4.5 in Sections 4.3.

A.1. **Proof of Lemma 2.1.**

Proof. The fractional-power resolvent identity [8, formula (5.28)] gives

$$\mathcal{R}_{\alpha,0}(z) = \frac{z^{2-2\alpha}}{\alpha} R_0(z) + J_\alpha(z). \tag{A.1}$$

The Yukawa kernel of $(\gamma - \Delta)^{-1}$ is $e^{-|x|\sqrt{\gamma}}/(4\pi|x|)$, giving (2.1)–(2.2). To estimate the fractional correction and prove its holomorphy, set

$$D_\alpha(\gamma, z) = (z^{2\alpha} - \gamma^\alpha e^{i\pi\alpha})(z^{2\alpha} - \gamma^\alpha e^{-i\pi\alpha}).$$

On a closed subsector of the right half-plane,

$$|D_\alpha(\gamma, z)| \geq c(\gamma^{2\alpha} + |z|^{4\alpha}). \tag{A.2}$$

Indeed, the argument of $z^{2\alpha}$ stays strictly between $-\pi\alpha$ and $\pi\alpha$. Scaling γ by $|z|^2$ and compactness of the angular interval give the uniform bound.

The Fourier multiplier of $J_\alpha(z)$ is

$$b_\alpha(q,z) = \frac{\sin(\pi\alpha)}{\pi}\int_0^\infty \frac{\gamma^\alpha}{(\gamma+q^2)D_\alpha(\gamma,z)}\,\mathrm{d}\gamma, \qquad q=|\xi|.$$

Setting $a=|z|$ and $Q=q/a$, (A.2) yields

$$|b_\alpha(q,z)| \le Ca^{-2\alpha}\int_0^\infty \frac{s^\alpha}{(s+Q^2)(1+s^{2\alpha})}\,\mathrm{d}s \le C(a^2+q^2)^{-\alpha}.$$

For $Q\le 1$, use $s^{\alpha-1}/(1+s^{2\alpha})$ as a majorant. For $Q>1$, split the integral at Q^2; both resulting terms are $O(Q^{-2\alpha})$. This proves (2.4) by Plancherel. The same bound on compact subsets, together with the scalar Cauchy formula, proves holomorphy in the operator norm $L^2\to H^{2\alpha}$.

The classical localized resolvent is entire and satisfies

$$\|\chi_1 R_0(z)\chi_2\|_{L^2\to H^l} \le C|z|^{l-1}e^{T(\Im z)_-}, \qquad 0\le l\le 2,$$

for $|z|\ge 1$; see [3]. Combining this estimate with (A.1) gives (2.3).

The kernel is locally integrable, with leading singularity $c_\alpha|x|^{2\alpha-3}$ at the origin. At infinity, its classical part satisfies (1.2), while derivatives of the fractional correction have decay $O(|x|^{-3-2\alpha-j})$ at derivative order j. Thus convolution with compactly supported L^2 data is outgoing. For uniqueness, let w be a homogeneous solution in the radiating tempered class with Fourier transform locally integrable near zero. Then $(|\xi|^{2\alpha}-k^{2\alpha})\widehat{w}=0$. Its Fourier support is on $|\xi|=k$. Multiplication there by the smooth quotient $(|\xi|^2-k^2)/(|\xi|^{2\alpha}-k^{2\alpha})$ gives $(-\Delta-k^2)w=0$. Classical outgoing uniqueness gives $w=0$. This agrees with the limiting absorption construction in [11]. □

A.2. **Proof of Theorem 2.2.**

Proof. Choose $\chi\in C_c^\infty$ equal to one near $\operatorname{supp} V$, and consider

$$T_V(z) = I + \chi\mathcal{R}_{\alpha,0}(z)V \quad \text{on } L^2(\mathbb{R}^3).$$

By Lemma 2.1, $T_V(z)-I$ is holomorphic and compact: its range is supported in $\operatorname{supp}\chi$ and gains $2\alpha>0$ derivatives. If z is in the first quadrant and $T_V(z)v=0$, put $w=-\mathcal{R}_{\alpha,0}(z)Vv$. Here the spectral resolvent maps L^2 into $H^{2\alpha}$ globally. Since $\chi w=v$ and $Vw=Vv$, we have

$$((-\Delta)^\alpha + V - z^{2\alpha})w = 0.$$

Taking the imaginary part of its L^2 pairing with w gives $-\Im(z^{2\alpha})\,\|w\|_{L^2}^2=0$, hence $v=w=0$. The index-zero Fredholm operator is therefore invertible. Analytic Fredholm theory gives a meromorphic inverse on $\Re z>0$; let Σ_V be its positive real singularities. For a real frequency outside the exceptional set, set

$$v = T_V(k)^{-1}(\chi e^{ikx\cdot d}), \qquad u = e^{ikx\cdot d} - \mathcal{R}_{\alpha,0}(k)Vv.$$

Then $\chi u=v$, proving existence and uniqueness. The free mapping property gives local $H^{2\alpha}$ regularity and continuous dependence, and defines the localized perturbed resolvent.

Each exceptional frequency corresponds to an L^2 eigenfunction. Suppose $v \neq 0$ and $T_V(k)v = 0$, and set $w = -\mathcal{R}_{\alpha,0}(k)Vv$, $f = Vw = Vv$. The function f is compactly supported and in L^2, and

$$w = -\mathcal{R}_{\alpha,0}(k)f, \qquad \int_{\mathbb{R}^3} w\overline{f}\,\mathrm{d}x = \int_{\mathbb{R}^3} V|w|^2\,\mathrm{d}x \in \mathbb{R}.$$

The spectral boundary-value formula, paired with the compactly supported L^2 function f, gives

$$0 = \Im \int_{\mathbb{R}^3} (\mathcal{R}_{\alpha,0}(k)f)\overline{f}\,\mathrm{d}x = \frac{\pi}{(2\pi)^3} \int_{\mathbb{R}^3} \delta(|\xi|^{2\alpha} - k^{2\alpha})|\widehat{f}(\xi)|^2\,\mathrm{d}\xi. \tag{A.3}$$

Compact support makes $\widehat{f}$ continuously differentiable, so it vanishes on the spectral sphere. Thus the delta contribution in the outgoing multiplier vanishes, and the quotient $\widehat{f}(\xi)/(|\xi|^{2\alpha} - k^{2\alpha})$ is bounded in a neighborhood of that sphere. Away from the sphere the denominator is bounded away from zero on compact sets and has order 2α at infinity. It follows that $w \in H^{2\alpha}(\mathbb{R}^3)$ and

$$((-\Delta)^\alpha + V)w = k^{2\alpha}w. \tag{A.4}$$

Put $H_0 = (-\Delta)^\alpha$, $E = k^{2\alpha}$, and $U_a w(x) = a^{3/2}w(ax)$ for $a > 0$. The identity $H_0 U_a = a^{2\alpha}U_a H_0$ and self-adjointness give

$$(1 - a^{2\alpha})E\,\langle w, U_a w\rangle = \left\langle w, (V - a^{2\alpha}V(a\,\cdot))U_a w\right\rangle. \tag{A.5}$$

Dividing by $1 - a^{2\alpha}$ and letting $a \to 1$, strong continuity of U_a and uniform convergence of the potential difference quotient yield

$$E\,\|w\|_{L^2}^2 = \int_{\mathbb{R}^3} \left(V + \frac{x \cdot \nabla V}{2\alpha}\right)|w|^2\,\mathrm{d}x. \tag{A.6}$$

This contradicts $E > E_*(V)$ and proves (2.6). □

A.3. **Proof of Lemma 2.4.**

Proof. Set $h = k^{-1}$, $\delta = h^\mu$, and

$$H_h = |hD|^\mu + \delta V, \qquad A_h = \frac{1}{2}(x \cdot hD + hD \cdot x), \qquad D = -i\nabla.$$

Both operators are self-adjoint on their natural domains. The dilation group preserves $H^\mu(\mathbb{R}^3)$ and gives the exact commutator identity

$$[H_h, iA_h] = h\Big(\mu H_h - \delta(\mu V + x \cdot \nabla V)\Big). \tag{A.7}$$

For $j = 1, 2$, put $B_0 = H_h$ and $B_j = [B_{j-1}, iA_h]$. Then

$$B_j = h^j\Big(\mu^j |hD|^\mu + \delta(-x \cdot \nabla)^j V\Big), \qquad \Big\|B_j(H_h + i)^{-1}\Big\| \le C_j h^j. \tag{A.8}$$

These identities extend from a common core to operators from the domain of H_h, with its graph norm, to L^2. They follow from the exact dilation action on the Fourier multiplier, including at zero momentum.

Let $I = [3/4, 5/4]$ and denote the spectral projection of H_h onto I by $E_I(H_h)$. Since $\delta \to 0$, (A.7) implies, for all sufficiently small h,

$$E_I(H_h)[H_h, iA_h]E_I(H_h) \ge chE_I(H_h), \qquad c > 0.$$

The common core, commutator bounds, and strict lower bound verify the hypotheses of the semiclassical Mourre estimate in [9, Section 2, Theorem 1]. We apply the case $\beta = 1$, $n = 1$, and weight exponent $s = 1$, for which (H4) is unnecessary by the remark following the theorem. Its Lemmas 6–7 use only the first two commutator bounds for these parameters. The passage to spatial weights requires only a Lipschitz bound on V. Consequently, for λ in a fixed smaller interval about one,

$$\left\| \langle A_h \rangle^{-1} (H_h - \lambda - i0)^{-1} \langle A_h \rangle^{-1} \right\| \le C h^{-1}. \tag{A.9}$$

We obtain spatial weights from (A.9) by spectral localization. Choose a real $\varphi \in C_c^\infty(I)$ which is one on a neighborhood of the smaller energy interval. In the semiclassical Sobolev norm $\|v\|_{H_h^s} = \|\langle hD \rangle^s v\|_2$, we first show that

$$\|\varphi(H_h)\|_{L^2 \to H_h^1} \le C. \tag{A.10}$$

Indeed, multiplication by a uniformly Lipschitz V is bounded on H_h^s for $0 \le s \le 1$, uniformly for $0 < h \le 1$, and

$$\|v\|_{H_h^{s+\mu}} \le C_s \Big(\||hD|^\mu v\|_{H_h^s} + \|v\|_{H_h^s} \Big), \qquad |hD|^\mu = H_h - \delta V.$$

Starting from the spectral theorem, iterate this estimate with compactly supported smooth spectral multipliers until the output has one derivative. The input orders remain below one, where the Lipschitz multiplication bound on V applies. Functional calculus and (A.8) also give $\|[A_h, \varphi(H_h)]\| \le Ch$.

Put $W = \langle x \rangle^{-1}$, and write M_f for multiplication by f. The ordering identity

$$A_h W = \sum_{j=1}^{3} hD_j M_{x_j/\langle x \rangle} + \frac{3ih}{2} M_{1/\langle x \rangle}$$

and (A.10) imply that $\varphi(H_h) A_h W$ is uniformly bounded. Commuting A_h past $\varphi(H_h)$ and taking adjoints yield

$$\|\langle A_h \rangle \varphi(H_h) W\| + \|W \varphi(H_h) \langle A_h \rangle\| \le C.$$

These bounds extend from a core by the closedness of A_h. They allow us to factor

$$\begin{aligned} W\varphi(H_h)(H_h - \lambda - i0)^{-1}\varphi(H_h)W = {} & \Big(W\varphi(H_h)\langle A_h \rangle \Big) \\ & \cdot \Big(\langle A_h \rangle^{-1} (H_h - \lambda - i0)^{-1} \langle A_h \rangle^{-1} \Big) \Big(\langle A_h \rangle \varphi(H_h) W \Big). \end{aligned}$$

Its norm is at most Ch^{-1} by (A.9). The complementary term $W(1-\varphi(H_h)^2)(H_h-\lambda-i0)^{-1}W$ is uniformly bounded by the spectral theorem. We have proved

$$\left\| W (H_h - \lambda - i0)^{-1} W \right\| \le C h^{-1}.$$

At $\lambda = 1$, multiplication by h^μ gives (2.8). The constants depend only on a C^2 bound. Mollification and continuity of the localized Fredholm inverse therefore extend the estimate from smooth potentials to the stated class. □

A.4. **Proof of Proposition 2.5.**

Proof. Write the free splitting (A.1) as

$$\mathcal{R}_{\alpha,0}(z) = a(z)R_0(z) + J_z, \qquad a(z) = z^{2-\mu}/\alpha, \quad J_z = J_\alpha(z).$$

In a fixed closed proper sector of the right half-plane,

$$\|J_z\| \le C|z|^{-\mu}, \qquad \|\partial_z J_z\| \le C|z|^{-\mu-1}. \tag{A.11}$$

The derivative estimate follows from (2.4) and the Cauchy formula on discs of radius $c|z|$ in a slightly larger sector. For $\Re z$ large, define

$$B_z = I + J_z V, \quad C_z = I + V J_z, \quad W_z = C_z^{-1} V = V B_z^{-1}, \quad Q_z = a(z) W_z. \tag{A.12}$$

The inverses are uniformly bounded on L^2 by a Neumann series. If $K = \overline{B_r}$, then

$$W_z = 1_K W_z 1_K, \qquad \|W_z\| \le C, \qquad \|Q_z\| \le C|z|^{2-\mu}.$$

Here 1_K denotes multiplication by the characteristic function of K. The support property follows by expanding $W_z = V - V J_z V + V J_z V J_z V - \cdots$. At a real frequency k, J_k is self-adjoint, so W_k and Q_k are self-adjoint as well.

Set $S_z = (-\Delta - z^2 + Q_z)^{-1}_{\mathrm{out}}$. The outgoing resolvents satisfy

$$\begin{aligned} \mathcal{R}_{\alpha,V}(z) &= B_z^{-1} J_z + a(z) B_z^{-1} S_z C_z^{-1}, \\ a(z) S_z &= B_z \mathcal{R}_{\alpha,V}(z) C_z - J_z C_z. \end{aligned} \tag{A.13}$$

To verify them, let $u = \mathcal{R}_{\alpha,0}(z)(f - Vu)$ and set $w = B_z u - J_z f$. Direct substitution gives

$$w = a(z) R_0(z)(C_z^{-1} f - W_z w) = a(z) S_z C_z^{-1} f,$$

which proves (A.13) wherever the inverses exist; the identities then extend meromorphically.

For any fixed cutoff η, the second identity and Lemma 2.4 imply

$$\|\eta S_k \eta\| \le C k^{-1}. \tag{A.14}$$

Expanding $\eta B_k \mathcal{R}_{\alpha,V}(k) C_k \eta$ leaves only compactly localized occurrences of $\mathcal{R}_{\alpha,V}(k)$, since the adjacent factors are either η or V. The remaining factors J_k are bounded on global L^2. Division by $a(k)$ changes the power $k^{1-\mu}$ to k^{-1}, while the last term in (A.13) contributes $O(k^{-2})$.

To continue the estimate off the real axis, first freeze the perturbation at Q_k. Write

$$P^{(k)} = -\Delta + Q_k, \qquad S^{(k)}(z) = (P^{(k)} - z^2)^{-1}_{\mathrm{out}}.$$

This is a self-adjoint bounded perturbation of $-\Delta$ with domain H^2. Its cutoff resolvent is meromorphic in z by analytic Fredholm theory, since the perturbation has input and output supported in K and the localized classical free resolvent gains two derivatives. At $z = k$, it agrees with S_k.

Choose real cutoffs $\rho, \eta, \chi \in C_c^\infty(\mathbb{R}^3)$ with $\rho = 1$ near K, $\eta = 1$ near $\operatorname{supp}\rho$, and $\chi = 1$ near $\operatorname{supp}\eta$. Put

$$C_\rho = [-\Delta, \rho], \qquad F_z = \eta S^{(k)}(z) \eta, \qquad D_{z,k} = \chi(R_0(z) - R_0(k))\chi.$$

The two-sided support of Q_k gives $Q_k(1-\rho) = (1-\rho)Q_k = 0$. The cutoff resolvent identity of [3, Lemma 6.27] therefore takes the form

$$\begin{aligned} F_z - F_k = (z^2-k^2)F_z\rho(2-\rho)F_k \\ + \Big(\eta(1-\rho) + F_zC_\rho\Big)D_{z,k}\Big((1-\rho)\eta - C_\rho F_k\Big). \end{aligned} \tag{A.15}$$

The identities

$$\begin{aligned} S^{(k)}(z)(1-\rho) &= (1-\rho+S^{(k)}(z)C_\rho)R_0(z), \\ (1-\rho)S^{(k)}(k) &= R_0(k)(1-\rho-C_\rho S^{(k)}(k)) \end{aligned}$$

follow by applying the respective differential operators. Substituting $1 = \rho(2-\rho) + (1-\rho)^2$ into the ordinary resolvent identity and using these formulas yields (A.15).

In the norms $H^s_{1/k}$, the classical free estimate and Cauchy's formula on a fixed disc give

$$\|D_{z,k}\|_{H^{-r_1}_{1/k}\to H^{r_2}_{1/k}} \le C|z-k|k^{-1}, \quad r_1, r_2 \in \{0,1\}, \quad |z-k|<1; \tag{A.16}$$

see also [3, equation (6.5.29)] after rescaling. Moreover, C_ρ has norm $O(k)$ both from $H^1_{1/k}$ to L^2 and from L^2 to $H^{-1}_{1/k}$. Let $X = \|F_z\|$ and $Y = \|F_k\| \le C/k$. Expanding (A.15) and applying the commutators to the free difference give

$$X \le \frac{C}{k} + C|z-k|\left(X + \frac{1}{k}\right).$$

For the two-commutator term, $\|C_\rho D_{z,k} C_\rho\| \le Ck|z-k|$, so its contribution is at most $Ck|z-k|XY \le C|z-k|X$. Thus there is a fixed $b_0 > 0$ such that

$$\left\|\eta S^{(k)}(z)\eta\right\| \le Ck^{-1}, \qquad |z-k| < b_0. \tag{A.17}$$

The estimate holds away from the discrete poles and makes each pole in the disc removable. The constants depend only on the cutoffs and the uniform real-frequency bound in (A.14).

We now restore the frequency dependence of the perturbation. Differentiating (A.12) and using (A.11), we obtain

$$\|\partial_z W_z\| \le Ck^{-\mu-1}, \qquad \|Q_z - Q_k\| \le C|z-k|k^{1-\mu} \quad (|z-k|<b_0).$$

This difference is supported in K on both sides. Its product with the localized frozen resolvent has norm at most $C|z-k|k^{-\mu}$ by (A.17). For k large, a Neumann series therefore gives a holomorphic S_z on this disc with the same cutoff bound C/k. Finally, use $B_z^{-1} = I - J_zW_z$ and $C_z^{-1} = I - W_zJ_z$ in (A.13). All occurrences of S_z are compactly localized, and the remaining factors are uniformly bounded on L^2. This gives $\|\eta\mathcal{R}_{\alpha,V}(z)\eta\| \le Ck^{1-\mu}$. The continuations agree on overlapping discs, since they agree in the upper half-plane. Their union contains (2.9), after increasing k_0 and taking $b < b_0$. Arbitrary fixed cutoffs follow by choosing η to equal one on their supports. □

A.5. **Proof of Corollary 2.6.**

Proof. The source $Ve^{izx\cdot d}$ is holomorphic with values in $L^2(B_r\times\mathbb{S}^2)$ and is uniformly bounded in a fixed strip. By Proposition 2.5 and

$$u^s_V(\cdot,d,z) = -\mathcal{R}_{\alpha,V}(z)(Ve^{izx\cdot d}),$$

the scattered field is holomorphic on every fixed bounded volume observation set, with norm at most $C|z|^{1-\mu}$. In particular, (2.11) holds for $\mathcal{O} = A$, and

$$\|u_V(\cdot,\cdot,z)\|_{L^2(B_r\times\mathbb{S}^2)} \le C(1+|z|^{1-\mu}).$$

For $\mathcal{O} = \Gamma$, we use the separation $R - r$ to estimate the free kernel directly. Equations (2.2) and (A.2) give, for $a > 0$,

$$|J_{\alpha,z}(a)| \le \frac{C|z|^{-2\mu}}{a}\int_0^\infty \gamma^{\mu/2} e^{-a\sqrt{\gamma}}\,\mathrm{d}\gamma \le C|z|^{-2\mu}a^{-3-\mu}.$$

Consequently, the operator

$$T_\Gamma(z)f(x) = \int_{B_r} G_{\alpha,z}(x-y)f(y)\,\mathrm{d}y, \qquad x\in\Gamma,$$

is holomorphic from $L^2(B_r)$ to $L^2(\Gamma)$ and satisfies

$$\|T_\Gamma(z)\| \le C(|z|^{2-\mu} + |z|^{-2\mu}).$$

The kernel is holomorphic and uniformly bounded on the separated compact product for z in compact subsets of the strip, which gives holomorphy in the Hilbert–Schmidt norm. The identity $u_V^s|_\Gamma = -T_\Gamma(z)(Vu_V)$ proves the assertion, since $2-\mu+(1-\mu)_+ < 3$. Finally, the incident and modulating exponentials are bounded holomorphic multipliers on either fixed observation set. □

A.6. **Proof of Proposition 4.5.** Write $H_V = (-\Delta)^\alpha + V$ and $\mu = 2\alpha$. We first establish a spectral estimate for the spherical trace.

Lemma A.1. *Let $V \in W^{1,\infty}(\mathbb{R}^3;\mathbb{R})$, let $\lambda \notin \sigma(H_V)$, and put $t = \|(H_V-\lambda)^{-1}\|_{L^2\to L^2}$. Then*

$$\left\|(H_V-\lambda)^{-1}f\right\|_{H^1} \le C\left(t\,\|f\|_{H^1} + t^2\,\|\nabla V\|_\infty\,\|f\|_{L^2}\right). \tag{A.18}$$

The resolvent is holomorphic on H^1 as a function of λ in the spectral resolvent set.

Proof. Set $w = (H_V-\lambda)^{-1}f$. For translation T_a in the jth coordinate and $D_a = (T_a - I)/a$, translation invariance of the fractional Laplacian gives

$$(H_V-\lambda)D_a w = D_a f - (D_a V)T_a w.$$

This identity holds in the natural domain H^μ, which is preserved by translations. Thus

$$\|D_a w\|_{L^2} \le t\,\|\partial_j f\|_{L^2} + t^2\,\|\partial_j V\|_\infty\,\|f\|_{L^2}.$$

The difference-quotient characterization of H^1 proves (A.18). The resolvent identity and these locally uniform bounds prove holomorphy on H^1. □

Proof of Proposition 4.5. Let b_0 be the strip width in Proposition 2.5. Choose $0 < b < b_0/4$ and $0 < \epsilon < \pi/4$ so that $\mu(\pi+\epsilon) < 2\pi$. Increase L so that the domain avoids the chosen cut, $|z| \ge L - b > 1$, and

$$\mu\arg z \in [\mu\pi/2, \mu(\pi+\epsilon)] \Subset (0,2\pi) \quad \text{when } z\in\mathcal{D}_{b,L},\ \Re z \le 0.$$

Since H_V is self-adjoint and $\sigma(H_V)\subset[-F,\infty)$, a further increase of L gives

$$\left\|(H_V - z^\mu)^{-1}\right\|_{L^2\to L^2} \le C|z|^{-\mu}, \qquad \Re z\le 0, \quad z\in\mathcal{D}_{b,L}. \tag{A.19}$$

Indeed, the distance of $e^{i\theta}$ from $[0,\infty)$ has a positive minimum on the displayed angular interval, and

$$\operatorname{dist}(z^\mu,[-F,\infty)) \geq c_\mu|z|^\mu - F \geq \tfrac{1}{2}c_\mu|z|^\mu.$$

In the first quadrant, $\sin(\mu\theta) \geq c_\mu \sin\theta$ for $0<\theta\leq\pi/2$, and hence

$$\left\|(H_V - z^\mu)^{-1}\right\|_{L^2\to L^2} \leq \frac{1}{\Im(z^\mu)} \leq \frac{C|z|^{1-\mu}}{\Im z}. \tag{A.20}$$

Use the spectral resolvent on the two open sets $\mathcal{D}_{b,L}\cap\{\Im z>0\}$ and $\mathcal{D}_{b,L}\cap\{\Re z<0\}$, and the outgoing resolvent on

$$\mathcal{D}_{b,L}\cap\{\Re z > k_c,\ \Im z < 2b\},$$

where k_c is the lower endpoint of the fixed strip. Taking $L>\sqrt{k_c^2+9b^2}+1$ makes these sets an open cover of $\mathcal{D}_{b,L}$. The spectral and outgoing resolvents agree on their first-quadrant overlap and therefore define a holomorphic extension. The strip estimate, (A.19), and (A.20) for $\Im z\geq b$ prove (4.10).

On the spectral pieces, write

$$u_V^s(z) = -\big(H_V - z^\mu\big)^{-1} f_V(z), \qquad f_V(z,x,d) = V(x)e^{izx\cdot d}.$$

The source is entire with values in $H^1_x L^2_d$ and satisfies

$$\|f_V(z)\|_{L^2_{x,d}} \leq Ce^{r|\Im z|}, \qquad \|f_V(z)\|_{H^1_xL^2_d} \leq C(1+|z|)e^{r|\Im z|}.$$

On $\Re z\leq 0$ and on $\Re z\geq 0$, $\Im z\geq b$, the spectral resolvent norm is at most $C(1+|z|)$. Lemma A.1 consequently yields

$$\|u_V^s(z)\|_{H^1_xL^2_d} \leq C(1+|z|)^2 e^{r(\Im z)_+}. \tag{A.21}$$

Here $e^{r|\Im z|}\leq e^{rb}e^{r(\Im z)_+}$ in the domain. The bounded trace map $H^1(\mathbb{R}^3)\to L^2(\partial B_R)$ gives the spherical bound, and the L^2 restriction map gives the annular bound. These bounded maps and the H^1 resolvent holomorphy also give Y-valued holomorphy.

On the first-quadrant overlap, the compact-source free-kernel formula for this field is smooth at the receivers, since $R-r>0$. Its H^1 trace therefore equals the spherical restriction in Corollary 2.6. Hence the spectral traces and the outgoing receiver fields define one holomorphic continuation of the measured field. On the remaining bounded-height strip, Corollary 2.6 supplies the cubic bound. Combining it with (A.21) proves (4.11). □

Appendix B. Symbol and free propagation estimates

We prove the finite symbol expansions and free propagation bounds used in Section 3. Throughout this appendix, $\mu\in(0,2)$ is fixed.

B.1. **Finite expansions of the smooth symbol.** Use the symbol p_{sm} chosen in Section 3.1, and let $K=\mathcal{F}^{-1}p_{\mathrm{sm}}$. Away from zero, its kernel satisfies

$$|K(v)|\leq C|v|^{-3-\mu} \quad (0<|v|\leq 1), \qquad |K(v)|\leq C_M|v|^{-M} \quad (|v|\geq 1) \tag{B.1}$$

for every sufficiently large integer M. The first bound follows from the finite-part kernel of $|D|^\mu$, since $|\xi|^\mu - p_{\mathrm{sm}}$ is compactly supported and integrable. The second follows by integration by parts in frequency, since sufficiently high derivatives of p_{sm} are integrable. The

finite-part subtraction at zero involves only the value and first derivatives of the function on which the kernel acts. For a fixed radial cutoff ϑ equal to one near zero,

$$\langle K, F\rangle = c_* F(0) + \int K(v)\Big(F(v) - \vartheta(v)F(0) - \vartheta(v)v\cdot\nabla F(0)\Big)\,\mathrm{d}v.$$

Here c_* depends only on the fixed symbol and cutoff; radial symmetry removes first-order terms supported at zero, and the order $\mu < 2$ excludes higher-order ones. On a function vanishing to second order at zero, it acts by ordinary integration against K.

Let S be a real phase with bounded momentum $\Xi = \nabla S$, and let a have bounded derivatives of the indicated orders. Set

$$F_t(x,v) = e^{i\Phi_t(x,v)}a(x-tv), \qquad \Phi_t(x,v) = -\int_0^1 v\cdot\Xi(x-stv)\,\mathrm{d}s.$$

Then

$$e^{-iS(x)/h}p_{\mathrm{sm}}(hD)(e^{iS/h}a)(x) = \langle K, F_h(x,\cdot)\rangle.$$

If $a, \Xi \in C^L$ and $L \geq 2$, direct differentiation gives

$$|\partial_t^L F_t(x,v)| \leq C|v|^L(1+|v|^L).$$

Taylor's formula through degree $L-1$ has a remainder $\mathcal{E}_{L,h}$ bounded by $Ch^L|v|^L(1+|v|^L)$. Moreover,

$$F_t(x,0) = a(x), \qquad \nabla_v F_t(x,0) = -i\Xi(x)a(x) - t\nabla a(x),$$

so $\mathcal{E}_{L,h}$ and its first v derivatives vanish at zero. Since $L \geq 2 > \mu$, (B.1) yields

$$|\langle K, \mathcal{E}_{L,h}\rangle| \leq Ch^L \int |K(v)|\,|v|^L(1+|v|^L)\,\mathrm{d}v \leq Ch^L. \tag{B.2}$$

Thus

$$e^{-iS/h}p_{\mathrm{sm}}(hD)(e^{iS/h}a) = \sum_{\ell=0}^{L-1} h^\ell \mathscr{L}_{\ell,S}a + h^L r_{L,h}, \qquad \|r_{L,h}\|_\infty \leq C. \tag{B.3}$$

The expansion uses L derivatives of a and Ξ; differentiation in x through order q requires $L+q$ derivatives. If $a = O_{C^L}(\delta)$, the remainder is $O(\delta)$. The same conclusion holds when $a-1$ and $\Xi - d$ are $O_{C^L}(\delta)$, since every positive t derivative then contains a differentiated amplitude or a positive spatial derivative of Ξ.

The bilinear phase representation also gives the coefficients. With $A_x(z) = \int_0^1 \Xi(x+tz)\,\mathrm{d}t$, they are

$$\mathscr{L}_{\ell,S}a(x) = \frac{1}{i^\ell}\sum_{|\beta|=\ell}\frac{1}{\beta!}\,\partial_z^\beta\Big((\partial_\xi^\beta p_{\mathrm{sm}})(A_x(z))a(x+z)\Big)\Big|_{z=0}. \tag{B.4}$$

The change of variables $\eta = \xi - A_x(z)$ reduces the phase to $-z\cdot\eta$, whose diagonal Taylor coefficients give (B.4). The coefficient of a derivative of order q of a uses derivatives of Ξ up to order $\ell - q$. In particular,

$$\mathscr{L}_{0,S}a = p_{\mathrm{sm}}(\Xi)a, \qquad \mathscr{L}_{1,S}a = i^{-1}\mathcal{T}_S a, \qquad \mathcal{T}_S a = v_S\cdot\nabla a + \tfrac{1}{2}(\operatorname{div} v_S)a,$$

where $v_S = \nabla_\xi p_{\mathrm{sm}}(\Xi)$.

For a fixed smooth cutoff χ, the commutator has the representation

$$e^{-iS/h}[p_{\mathrm{sm}}(hD),\chi](e^{iS/h}a) = \langle K, e^{i\Phi_h}(\chi(x-hv)-\chi(x))a(x-hv)\rangle.$$

The identity

$$\chi(x-hv)-\chi(x) = -h\int_0^1 v\cdot\nabla\chi(x-thv)\,\mathrm{d}t$$

supplies one power of h before Taylor expansion. Consequently, for $L \geq 2$,

$$e^{-iS/h}[p_{\mathrm{sm}}(hD), \chi](e^{iS/h}a) = \sum_{\ell=1}^{L-1} h^{\ell}[\mathscr{L}_{\ell,S}, \chi]a + h^{L} r^{\chi}_{L,h}. \tag{B.5}$$

The C^q bound for $r^{\chi}_{L,h}$ requires $L - 1 + q$ derivatives of a and Ξ. The Taylor remainder vanishes to order L in v, making its kernel integral convergent at zero. If $a = O_{C^{L-1+q}}(\delta)$, this remainder is $O_{C^q}(\delta)$.

These expansions hold uniformly on the fixed construction sets. The phase may be extended by a fixed cutoff applied to $S - x \cdot d$, and amplitudes by cutoffs applied to their deviations from their incoming values. For small δ, the extensions retain $|\nabla S| \geq 1/2$ and the stated finite derivative bounds. Outside a fixed neighborhood of the spatial support of an input, (B.1) gives arbitrarily small powers of h with any fixed polynomial spatial weight. The same holds for a commutator outside a fixed neighborhood of the cutoff derivative support, where its kernel is evaluated at a fixed positive separation.

B.2. **Separated source and receiver estimates.** We use the decomposition

$$R_{h,0} = \frac{2}{\mu}(-h^2\Delta - 1 - i0)^{-1} + j_{\mu}(hD), \qquad j_{\mu}(\xi) = \frac{1}{|\xi|^{\mu} - 1} - \frac{2/\mu}{|\xi|^2 - 1}.$$

The convolution kernel J_h of the second term is positive, and the Yukawa representation gives

$$\|J_h\|_1 = \frac{2}{\mu} - 1, \qquad |\partial^{\beta} J_h(z)| \leq C_{\beta} h^{\mu} |z|^{-3-\mu-|\beta|} \quad (|z| \geq c > 0). \tag{B.6}$$

For sources in the potential support,

$$\|R_{h,0}\|_{L^2(B_r) \to L^2(\Gamma)} \leq C h^{-1}. \tag{B.7}$$

Parameterize a receiver patch by $X(x')$, $x' \in \mathbb{R}^2$, and put $\varphi(x', y) = |X(x') - y|$. Fix $r < r_1 < R$, extend the source by zero, and insert a smooth source cutoff equal to one on B_r and supported in B_{r_1}. With $\omega = (X - y)/|X - y|$, its mixed derivative matrix is

$$\partial_{x'}\partial_y \varphi = -|X - y|^{-1}(D_{x'}X)^T(I - \omega \otimes \omega).$$

This matrix has rank two, since $\omega \cdot n_X \geq (R - r_1)/(R + r_1) > 0$ on the cutoff support. After a finite partition, write $y = (y', t)$ so that the corresponding two by two minor is uniformly invertible. For fixed t, let $T_{h,t}$ have kernel $e^{i\varphi(x', y', t)/h} a(x', y', t)$, where a is the smooth compactly supported geometric amplitude. On sufficiently small patches,

$$|\nabla_{y'}(\varphi(x', y', t) - \varphi(z', y', t))| \geq c|x' - z'|.$$

Integration by parts bounds the kernel of $T_{h,t}T^*_{h,t}$ by $C_N(1 + |x' - z'|/h)^{-N}$ for any fixed N. Schur's test in two dimensions yields $\|T_{h,t}\| \leq Ch$, uniformly in t. Integration in t and Cauchy–Schwarz give the same bound from the source volume to the receiver patch. The factor h^{-2} in the classical kernel proves its contribution to (B.7). The separated bound in (B.6) controls the remaining term by Ch^{μ}.

The same estimate holds with any fixed source radius smaller than R.

B.3. **Proof of Lemma 3.2.**

Proof. Rotate coordinates so that $d = e_3$, and write $\xi = (\eta, \tau)$, with $\eta \in \mathbb{R}^2$. On the support of q_d, put

$$a_+(\eta) = \sqrt{1 - |\eta|^2}, \qquad b(\eta, \tau) = \frac{(|\eta|^2 + \tau^2)^{\mu/2} - 1}{\tau - a_+(\eta)}.$$

The quotient extends smoothly across $\tau = a_+(\eta)$ and is positive and bounded away from zero on a fixed neighborhood of $(0, 1)$. Define $m = q_d/b$ on this neighborhood and extend it by zero outside the support of q_d. The outgoing multiplier factors as

$$R_{h,0} q_d(hD) = (hD_t - a_+(hD_p) - i0)^{-1} m(hD). \tag{B.8}$$

Here a_+ may be extended to a smooth real function outside a fixed small neighborhood of the origin. Choose $\psi \in C_c^\infty(\mathbb{R}^2)$ equal to one on the transverse frequency projection of $\operatorname{supp} m$, and supported in $|\eta| < 2\varepsilon$. For $G = m(hD)f$, the retarded formula is

$$R_{h,0} q_d(hD) f(p, t) = \frac{i}{h} \int_{-\infty}^{t} e^{i(t-s)a_+(hD_p)/h} \psi(hD_p) G(\cdot, s)(p) \, \mathrm{d}s. \tag{B.9}$$

One-dimensional outgoing Fourier inversion in τ proves the identity for smooth compactly supported f; the estimates below extend it to the stated class.

Choose fixed radii

$$a < a_1 < b < c < R.$$

The portion of Γ with $|p| < c$ consists of the two graphs

$$t = \rho_\pm(p) = \pm\sqrt{R^2 - |p|^2}.$$

Let $\chi(p)$ be a smooth cutoff supported in $|p| < c$ and equal to one on $|p| \le b$. For either graph, consider the variable-time propagator

$$T_\rho v(p) = \chi(p)(2\pi h)^{-2} \int e^{i((p-z)\cdot\eta + \rho(p) a_+(\eta))/h} \psi(\eta) v(z) \, \mathrm{d}z \, \mathrm{d}\eta.$$

Its phase in (p, η) has mixed derivative matrix

$$I + \nabla\rho(p) \otimes \nabla a_+(\eta).$$

Since $\nabla\rho$ is bounded on $|p| \le c$, a sufficiently small ε keeps the determinant uniformly away from zero. The semiclassical Fourier transform is unitary, and the two-dimensional nondegenerate oscillatory integral estimate therefore gives

$$\|T_\rho\|_{L^2(\mathbb{R}^2) \to L^2(\mathbb{R}^2)} \le C. \tag{B.10}$$

After finite localization in (p, η), the kernel of the unnormalized oscillatory integral operator composed with its adjoint is bounded by $C_M(1 + |p - p'|/h)^{-M}$. Integration by parts in η gives this bound, since the gradient of the phase difference is at least $c|p - p'|$. Schur's test gives an $O(h)$ norm for the unnormalized two-dimensional operator, and its Fourier normalization cancels this factor.

Inserting $t = \rho(p)$ in (B.9) gives

$$\chi(p) R_{h,0} q_d(hD) f(p, \rho(p)) = \frac{i}{h} \int_{\mathbb{R}} \mathbf{1}_{s < \rho(p)} T_\rho e^{-is a_+(hD_p)/h} G(\cdot, s)(p) \, \mathrm{d}s.$$

For each s, the characteristic function multiplies the output and has L^2 operator norm at most one. The fixed-time propagator is unitary. Minkowski's inequality and (B.10) give

$$\|\chi R_{h,0} q_d(hD) f|_{t=\rho(p)}\|_2 \le Ch^{-1} \int_{\mathbb{R}} \|G(\cdot, s)\|_{L^2_p} \, \mathrm{d}s.$$

The kernel M_h of $m(hD)$ has a uniform $L^1(\mathbb{R}^3)$ bound, so mixed-norm Young's inequality yields

$$\|G\|_{L^1_s L^2_p} \le C\|f\|_{L^1_s L^2_p} \le C_T \|f\|_2.$$

The graph surface measure is uniformly comparable with dp, proving the estimate on both caps.

On the belt $|p| \ge b$, choose $T_1 > T$ and a fixed smooth cutoff χ_1 equal to one on a neighborhood of K_d, with

$$\operatorname{supp} \chi_1 \subset \{|p| < a_1, \ |s| < T_1\}.$$

Write $G = G_{\text{near}} + G_{\text{far}}$, where $G_{\text{near}} = \chi_1 G$. Since m is smooth and compactly supported, its kernel is Schwartz, and the positive distance between K_d and the support of $1 - \chi_1$ gives, for every M,

$$\|G_{\text{far}}\|_{L^1(\mathbb{R}^3)} \le C_M h^M \|f\|_2. \tag{B.11}$$

Split (B.9) accordingly, retaining $\psi(hD_p)$ on both terms. The propagator kernel with this cutoff is bounded in absolute value by Ch^{-2} for all time differences. Thus the far contribution to (B.9) is bounded pointwise by $C_M h^{M-3} \|f\|_2$.

For the near contribution, $|z| < a_1$ and $|s| < T_1$, while $x \in \Gamma$ gives $|t| \le R$. The transverse phase gradient in the propagator kernel is

$$p - z + (t - s)\nabla a_+(\eta).$$

Reduce ε so that, uniformly on the support of ψ,

$$(R + T_1)|\nabla a_+(\eta)| \le \tfrac{1}{2}(b - a_1).$$

Thus the gradient has magnitude at least $(b - a_1)/2$ on the belt. Repeated integration by parts in η bounds this kernel by $C_M h^{M-2}$ for any prescribed order. The time-ordering factor $\mathbf{1}_{s<t}$ is independent of η. Since $\|G_{\text{near}}\|_1 \le C\|G\|_2 \le C\|f\|_2$, the near contribution is at most $C_M h^{M-3}\|f\|_2$. Together with (B.11), this gives arbitrarily small powers of h on the belt and proves (3.9) on the whole sphere.

Finally suppose the source is downstream of the observation set as in (3.10). For every $x \in K_O$, the retarded formula uses only $s < t \le T_-$, which is separated from the source support $s \ge T_+$ by a fixed positive distance. Schwartz kernel decay therefore gives

$$\|\mathbf{1}_{s \le T_-} G\|_{L^1(\mathbb{R}^3)} \le C_M h^M \|f\|_2.$$

The absolute Ch^{-2} bound for the transversely localized propagator and the factor h^{-1} in (B.9) prove (3.10). The constants are uniform in d by rotation, since the geometric parameters and symbol bounds are fixed. □

Appendix C. A continuation estimate

The estimate below uses a fixed initial subinterval of the measurements, so its constants are independent of the upper measured frequency.

Lemma C.1. *Let $b > 0$, $L > b$, and $K_0 \geq 2L + 2$. Suppose that F is holomorphic with values in a complex Banach space $\mathcal{H}$ on*

$$\mathcal{D}_{b,L} = \{z : \Im z > -b,\ |z + ib| > L\}, \qquad \|F(z)\|_{\mathcal{H}} \leq 1.$$

If $\|F(k)\|_{\mathcal{H}} \leq \varepsilon$ on $[K_0, K_0 + 1/2]$, where $0 < \varepsilon < 1$, then

$$\|F(q)\|_{\mathcal{H}} \leq \exp\left(-\frac{c|\log\varepsilon|}{(1+q)^2}\right), \qquad q \geq K_0 + 1. \tag{C.1}$$

Here $c > 0$ depends only on b, L, K_0.

Proof. The Joukowski map

$$J(z) = \zeta + \frac{L^2}{\zeta}, \qquad \zeta = z + ib,$$

maps $\mathcal{D}_{b,L}$ conformally onto the upper half-plane $\mathbb{C}_+$. Indeed, $\Im J(z) = \Im\zeta(1 - L^2/|\zeta|^2) > 0$. The roots of $\zeta^2 - J(z)\zeta + L^2 = 0$ have product L^2, and the unique root outside $|\zeta| = L$ lies in the upper half-plane. This defines the holomorphic inverse.

Set $I = [K_0, K_0 + 1/2]$, $\eta(t) = J(t)$, and $\mathcal{A} = \eta(I)$. On this compact smooth arc,

$$\frac{3b}{4} \leq \Im\eta(t) \leq b, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\Re\eta(t) = 1 - L^2\frac{t^2 - b^2}{(t^2 + b^2)^2} \geq \frac{3}{4}. \tag{C.2}$$

Let

$$g(w, \eta) = \frac{1}{2\pi}\log\left|\frac{w - \overline{\eta}}{w - \eta}\right| = \frac{1}{4\pi}\log\left(1 + \frac{4\Im w\Im\eta}{|w - \eta|^2}\right), \qquad U(w) = \int_I g(w, \eta(t))\,\mathrm{d}t.$$

The positive potential U is harmonic off $\mathcal{A}$, bounded and continuous on $\mathbb{C}_+$, zero on the real boundary, and tends to zero at infinity. The monotonicity in (C.2) bounds the local logarithmic singularity by $C + |\log|t - t_0||$ with integrals over $|t - t_0| < h$ bounded by $Ch(1 + |\log h|)$, uniformly in w near the arc. Here t_0 is the inverse image of the projection of $\Re w$ onto the interval $\Re\eta(I)$. Splitting the integral at h proves continuity and boundedness. The Green function gives the boundary limits and decay.

Choose a fixed $B \geq \max(1, \sup U)$ and write $u = U/B$. For $\ell \in \mathcal{H}^*$ with $\|\ell\| \leq 1$, put

$$v(w) = \log|\ell(F(J^{-1}(w)))|, \qquad E = |\log\varepsilon|.$$

This function is subharmonic, bounded above by zero, and at most $-E$ on $\mathcal{A}$. Since $0 \leq u \leq 1$, the function $v + Eu$ has upper boundary limit at most zero on the arc, the real axis, and at infinity. The maximum principle on $\mathbb{C}_+ \setminus \mathcal{A}$ yields $v \leq -Eu$. On bounded exhaustions, the added circular boundary contributes at most $E\sup_{|w|=T} u(w) \to 0$.

For $q \geq K_0 + 1$, the point $w_q = J(q)$ satisfies $3b/4 \leq \Im w_q \leq b$, while $|w_q - \eta(t)|^2 \leq C(1 + q)^2$ for $t \in I$. The Green function therefore gives

$$U(w_q) \geq \frac{c_1}{(1+q)^2}.$$

It follows that $|\ell(F(q))| \leq \exp(-cE/(1 + q)^2)$. Taking the supremum over such ℓ proves (C.1). □

References

[1] G. Bao, J. Zhai, and Y. Zhao, Stability for the inverse potential scattering problems with multi-wavenumbers, SIAM J. Math. Anal., 58 (2026), 2111–2137.

[2] S. Das, T. Ghosh, and S. Ma, Inverse scattering for the fractional Schrödinger equation, arXiv:2509.12685 (2025), revised 2026.

[3] S. Dyatlov and M. Zworski, *Mathematical Theory of Scattering Resonances*, Graduate Studies in Mathematics, vol. 200, American Mathematical Society, Providence, RI, 2019.

[4] T. Ghosh, M. Salo, and G. Uhlmann, The Calderón problem for the fractional Schrödinger equation, Anal. PDE, 13 (2020), 455–475.

[5] N. Laskin, Fractional quantum mechanics, Phys. Rev. E, 62 (2000), 3135–3145.

[6] N. Laskin, Fractional quantum mechanics and Lévy path integrals, Phys. Lett. A, 268 (2000), 298–305.

[7] P. Li and Z. Li, An inverse random source problem for the fractional Helmholtz equation, arXiv:2602.19559 (2026).

[8] C. Martínez and M. Sanz, *The Theory of Fractional Powers of Operators*, North-Holland Mathematics Studies, vol. 187, Elsevier, Amsterdam, 2001.

[9] S. Nakamura, Resolvent estimates and time-decay in the semiclassical limit, Astérisque, 210 (1992), 247–262.

[10] D. Qiu, X. Xu, and Y. Zhao, Stability for the inverse source problem of the stochastic fractional Helmholtz equation, arXiv:2609.27383 (2026).

[11] D. Zilberberg, F. Cakoni, and M. S. Vogelius, Limiting absorption principle and radiation condition for the fractional Helmholtz equation, arXiv:2602.18387 (2026).

[12] N. S. Belevtsov and S. Yu. Lukashchuk, A fast algorithm for fractional Helmholtz equation with application to electromagnetic waves propagation, Appl. Math. Comput., 416 (2022), 126728.

[13] C. Glusa, H. Antil, M. D'Elia, B. van Bloemen Waanders, and C. J. Weiss, A fast solver for the fractional Helmholtz equation, SIAM J. Sci. Comput., 43 (2021), A1362–A1388.

[14] T. Ghosh, A. Rüland, M. Salo, and G. Uhlmann, Uniqueness and reconstruction for the fractional Calderón problem with a single measurement, J. Funct. Anal., 279 (2020), 108505.

[15] J. Jia, S. Yue, J. Peng, and J. Gao, Infinite-dimensional Bayesian approach for inverse scattering problems of a fractional Helmholtz equation, J. Funct. Anal., 275 (2018), 2299–2332.

[16] J. Kraisler and J. C. Schotland, Collective spontaneous emission and kinetic equations for one-photon light in random media, J. Math. Phys., 63 (2022), 031901.

[17] J. Kraisler and J. C. Schotland, Kinetic equations for two-photon light in random media, J. Math. Phys., 64 (2023), 111903.

[18] P. Li, X. Yao, and Y. Zhao, The scattering resonances for Schrödinger-type operators with unbounded potentials, SIAM J. Math. Anal., 56 (2024), 2149–2170.

[19] S. Liu, Y. Zhang, B. A. Malomed, and E. Karimi, Experimental realisations of the fractional Schrödinger equation in the temporal domain, Nat. Commun., 14 (2023), 222.

[20] M. Naber, Time fractional Schrödinger equation, J. Math. Phys., 45 (2004), 3339–3352.

[21] A. Rüland, On single measurement stability for the fractional Calderón problem, SIAM J. Math. Anal., 53 (2021), 5094–5113.

[22] V. E. Tarasov and J. J. Trujillo, Fractional power-law spatial dispersion in electrodynamics, Ann. Phys., 334 (2013), 1–23.

[23] Y. Zhang, X. Liu, M. R. Belić, W. Zhong, Y. Zhang, and M. Xiao, Propagation dynamics of a light beam in a fractional Schrödinger equation, Phys. Rev. Lett., 115 (2015), 180403.

[24] D. Zilberberg, Reconstruction of the support of an inhomogeneity for the fractional Helmholtz equation, arXiv:2604.26154 (2026).

[25] X. Li, A numerical method for reconstructing the potential in fractional Calderón problem with a single measurement, arXiv:2409.16711 (2024).

[26] M. Dwivedi, J. Railo, and A. Rupp, A Galerkin finite element method for the fractional Calderón problem, arXiv:2603.26287 (2026).

[27] Y.-L. Liu, J. Wen, and J.-L. Fu, An efficient method for reconstructing the potential in the fractional Calderón problem with a single measurement, arXiv:2610.34175 (2026).

[28] P.-Z. Kow, J. Nurminen, and J. Railo, Bayesian inference for the fractional Calderón problem with a single measurement, arXiv:2511.11068 (2025), revised 2026.

School of Mathematical Sciences, Zhejiang University, Hangzhou 310058, China

Email address: qiudong@zju.edu.cn

School of Mathematical Sciences, and Center for Interdisciplinary Applied Mathematics, Zhejiang University, Hangzhou 310058, China

Email address: xxu@zju.edu.cn

School of Mathematics and Statistics, and Key Lab NAA-MOE, Central China Normal University, Wuhan 430079, China

Email address: zhaoy@ccnu.edu.cn